4. Commercial Aircraft Operators



4. Commercial Aircraft Operators

4.1 Business Models and Networks

4.2 Operating Costs

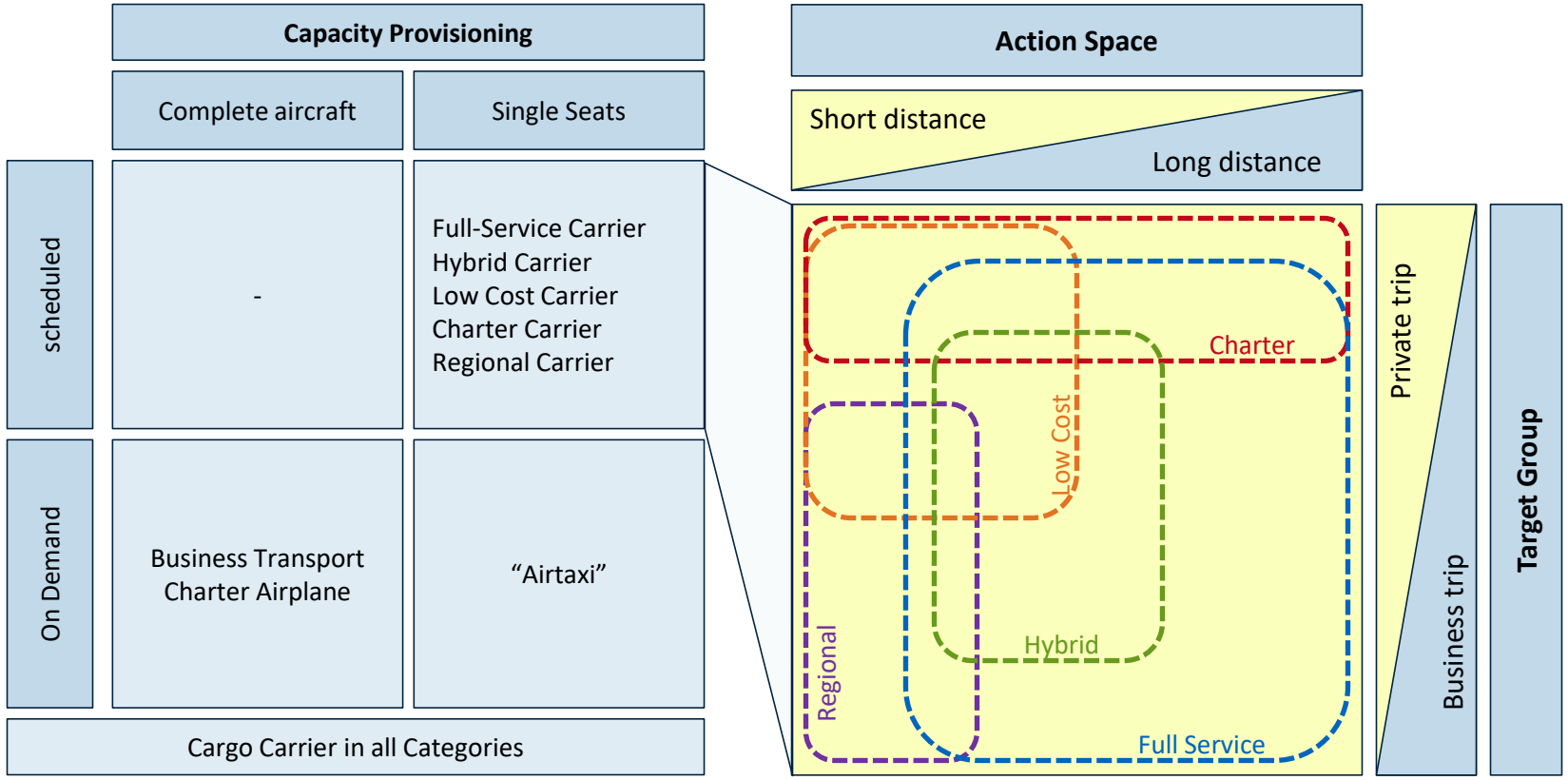
4.3 Airline Planning and Strategy

4. Commercial Aircraft Operators

4.1 Business Models and Networks



Air Operations Models



Based on Conrady (2013)

Airline Business Models

Criteria:

- Target Groups
- Revenue Models
  - ✓ Ticket Price
  - ✓ Other Sources of Revenues
- Network
  - ✓ Structure
  - ✓ Airports Served
- Aircraft Fleet
- Cooperation / Alliances
- …

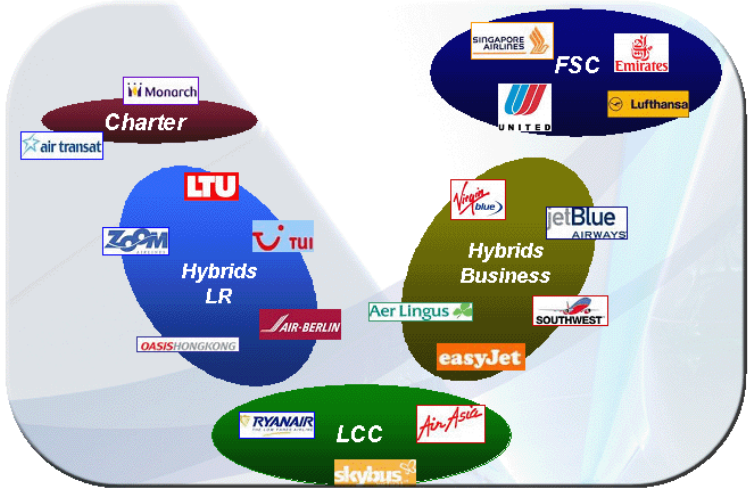
Business Cases:

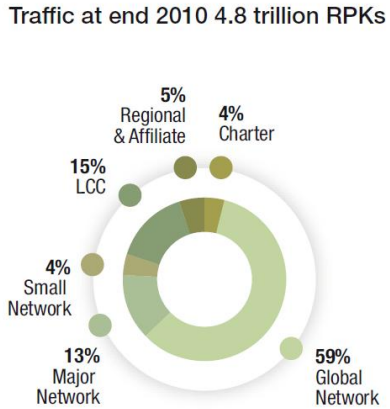
- Full-Service Network Carrier (FSC)
- Low-Cost Carrier (LCC)
- Charter / Holiday Carrier
- Regional Carrier
- Hybrid Carrier
- Cargo Carrier / Integrator

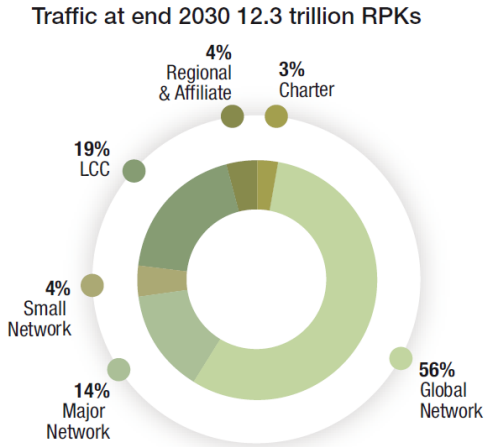
Source: DLR (2008)

Airline Business Models Changing

Dual-Brand Strategy	Long-Haul LCC	Hybrid/ No-Frills	Full-Service Carrier	Low-Cost Carrier
•Qantas – Jetstar •Singapore Airlines – Tiger Airways (LCC), Silkair (FSC regional) •Thai Airways – Thai Tiger (LCC), Thai Wings (FSC regional)	•Singapore Airlines – Scoot •Air Asia X •Norwegian •…	•jetblue •Azul •Southwest •Easyjet •…	•Lufthansa •British Airways •Air France •Singapore Airlines •United •…	•Ryanair •Air Asia •…



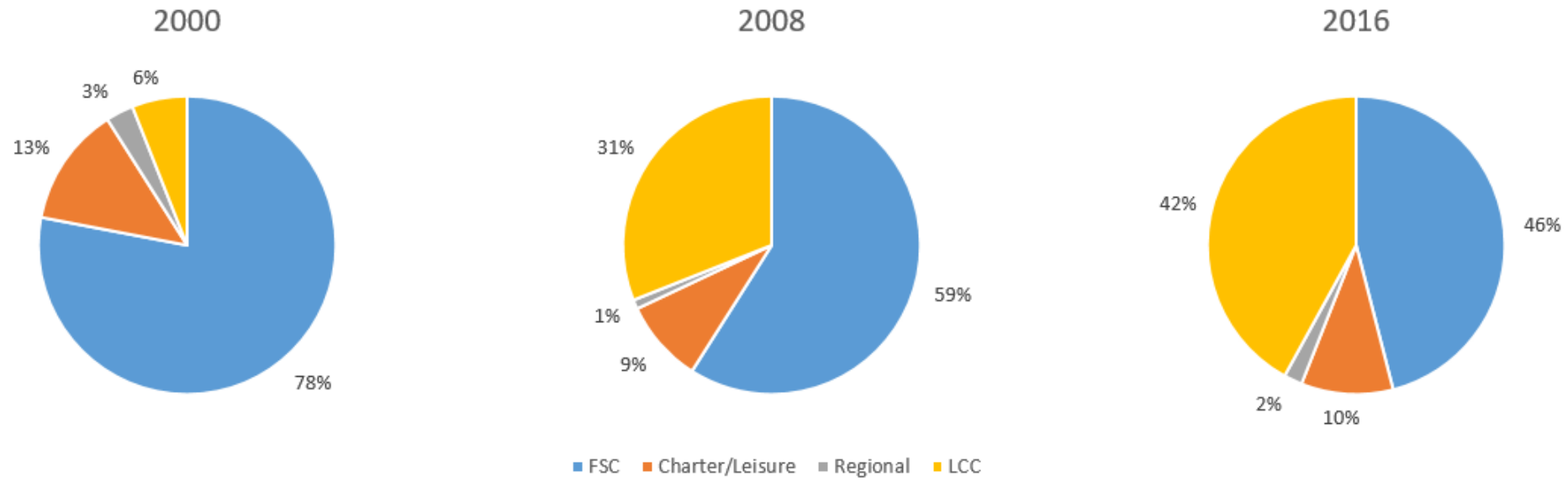




Source: Airbus 2011

Airline Business Models

Development of Market Shares in Europe



Source: OAG

Airline Business Models

Full-Service Carrier

	Examples	
Target Groups ➢ Business Travelers ➢ Frequent Travelers		
Revenue Generation ➢ Cost Coverage: Economy Class ➢ Earnings: Business & First Class		
Network ➢ Hub-and-Spoke Network ➢ Regional Feeder Traffic	Hubs: Frankfurt, Munich Feeder: LH Cityline	Hubs: Tokyo Haneda, Narita Feeder: Japan Air Commuter
Fleet ➢ Heterogeneous Structure ➢ Increasing Focus on Long-Range	Fleet (August 2018): Long-Range: **111** Short/Mid-Range: **175**	Fleet (August 2018): Long-Range: **114** Short/Mid-Range: **50**
Cooperations / Alliances ➢ Strong Global Alliances ➢ Star Alliance, Oneworld, SkyTeam	Star Alliance: 28 Airlines worldwide	Oneworld: 13 Airlines worldwide

Source: Airline Homepages, planespotters.net

Airline Business Models

Low-Cost Carrier

	Examples	
Target Groups ➢ Frequent Travelers ➢ Holiday Travelers	RYANAIR THE LOW FARES AIRLINE	Air Asia
Revenue Generation ➢ Cost Coverage: Economy Class ➢ Earnings: Ancillary Revenues		
Network ➢ Point-to-Point Network ➢ Mostly Short-/Mid-Range	Destinations: **216** Only Europe	Destinations: **26** International
Fleet ➢ Very Homogeneous Structure ➢ Frequently only 1 Type of Aircraft / Family	Fleet (August 2018): Boeing 737: **444**	Fleet (August 2018): Long-Range: **22**
Cooperations / Alliances ➢ FSC (sometimes) ➢ None	None	Air Asia & Virgin Group hold Shares

Source: Airline Homepages, planespotters.net

Airline Business Models

Charter / Holiday Carrier

	Examples	
		
Target Group ➢ Holiday Travelers		
Revenue Generation ➢ Cost Coverage: Economy Class ➢ Earnings: Ancillary Revenues		
Network ➢ Point-to-Point Network ➢ Short- / Mid- / Long-Range	Dest: **80** Europe and selected international dests.	Dest: over **109** Europe, Turkey, Asia
Fleet ➢ Homogeneous Fleet ➢ Sometimes, additional aircraft are leased	Fleet (August 2018): Long-Range: **29** Short/Mid-Range: **22**	Fleet (August 2018): Airbus A32S: **34** Boeing 737: **47**
Cooperations / Alliances ➢ Touristic Companies ➢ FSC	Owner: Thomas Cook (London, UK)	Owner: ESAS Holding (Turkish Holding Company)

Source: Airline Homepages, planespotters.net

Airline Business Models

Regional Carrier

	Examples	
	Lufthansa CityLine	JAPAN AIR COMMUTER
Target Groups ➢ Business Travelers ➢ Frequent Travelers		
Revenue Generation ➢ Cost Coverage: Economy Class ➢ Earnings: Ancillary Revenues		
Network ➢ Feeder within a Hub-and-Spoke Network ➢ Point-to-Point Network at Short-Range	Destinations: **80** Europe	Destinations: **14** Only within Japan
Fleet ➢ Homogeneous Structure ➢ Regional Jet Aircraft & Turboprops	Fleet (August 2018): CRJ: **35** ERJ: **14**	Fleet (August 2018): **15** (ATR 42,Dash 8, Saab 340)
Cooperations / Alliances ➢ FSC	100% Subsidiary of Lufthansa	Owner: Japan Air & Kagoshima Prefecture (Japanese State)

Source: Airline Homepages, planespotters.net

Airline Business Models

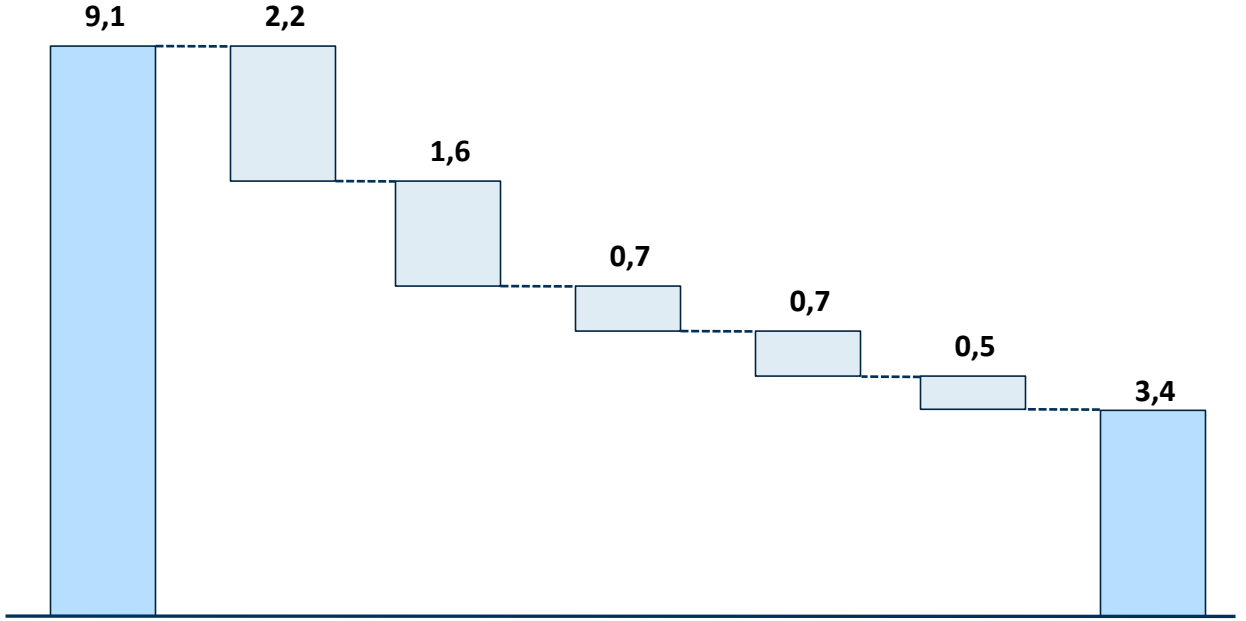
Hybrid Carrier

	Examples	
Target Group ➢ Large Range of Travelers		
Revenue Generation ➢ Cost Coverage: Economy Class ➢ Business/First Class, Ancillary Revenues	jetBlue	Aer Lingus
Network ➢ (Feeder within a) Hub-and-Spoke Network ➢ Point-to-Point Network at Short-/Long-Range	Hub: John F. Kennedy Airport, NY City	Hub: Dublin Europe,North America
Fleet ➢ Homogeneous Structure	Fleet: 193 Airbus A320/A321 60 Embraer 190	Fleet (August 2018): Long-Range: **17** Short/Mid Range: **37**
Cooperations / Alliances ➢ FSC ➢ Tourism Companies	Codeshare agreements with 21 airlines	Alliance: Oneworld

Source: Airline Homepages, planespotters.net

Saving Potentials of Low-Cost Carriers

Cost advantage of low-cost carriers vs. full-service carriers in € cents / ASK



Aircraft Operations	Airports	Distribution	Personnel	Service
• Denser seating • Higher utilization • Homogeneous fleet	• Smaller (cheaper) airports • Easier ground processes	• Internet distribution • No GDS* fees • No commissions • Spartan marketing & communication efforts	Lower costs • Cabin and crew • Administrative staff	Lower costs for air-to-ground services

Based on Conrady (2013)

*Global Distribution System

Hub-&-Spoke Networks

Multi-Hub System



Example: United Airlines (Domestic USA)



Hub-and-Spoke Systems

Advantages:

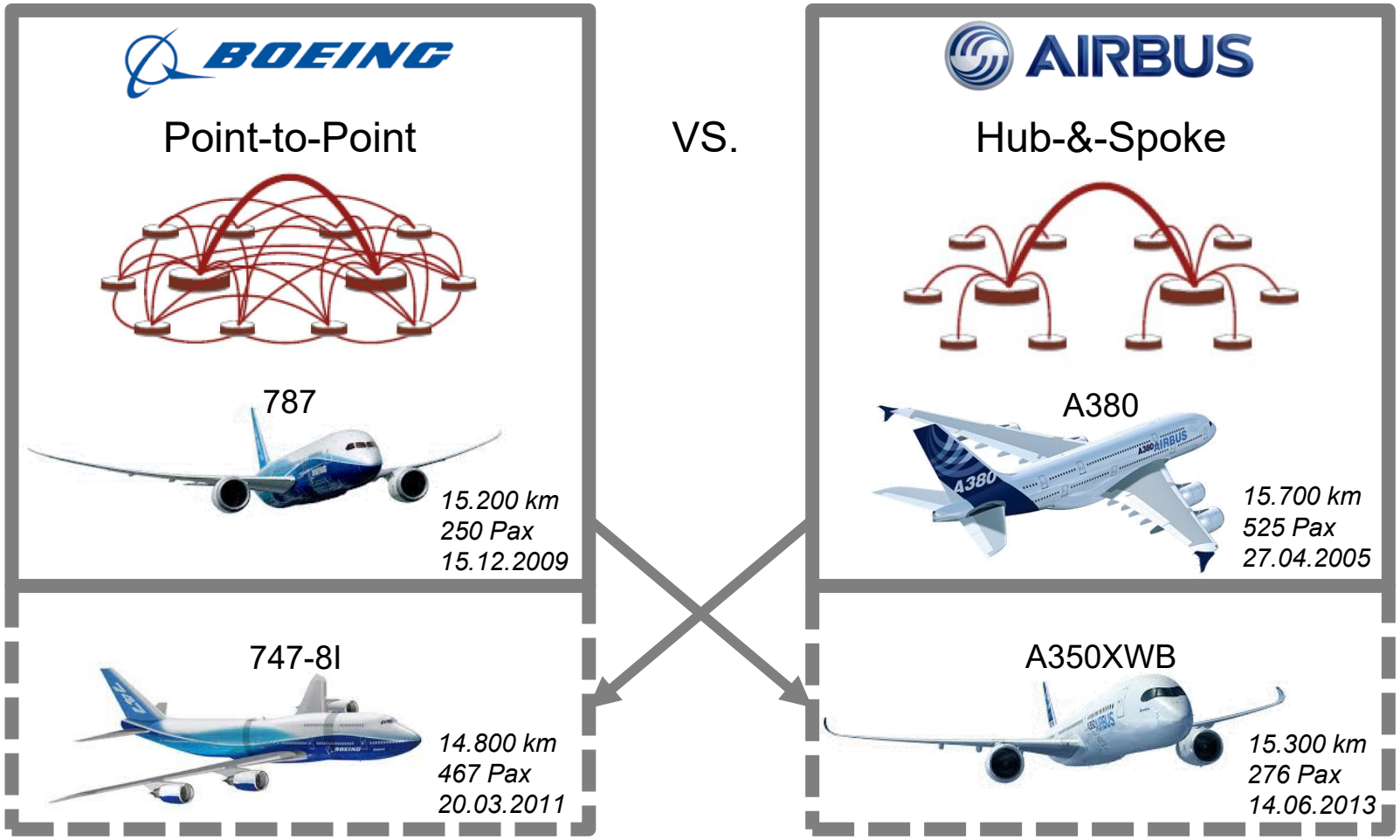
- Reduced costs by using larger aircraft
- Simplified planning of regular maintenance
- Connecting places with less original volume through the aviation network
- Simplified planning and provision of aircraft, personnel, spare parts

Disadvantages:

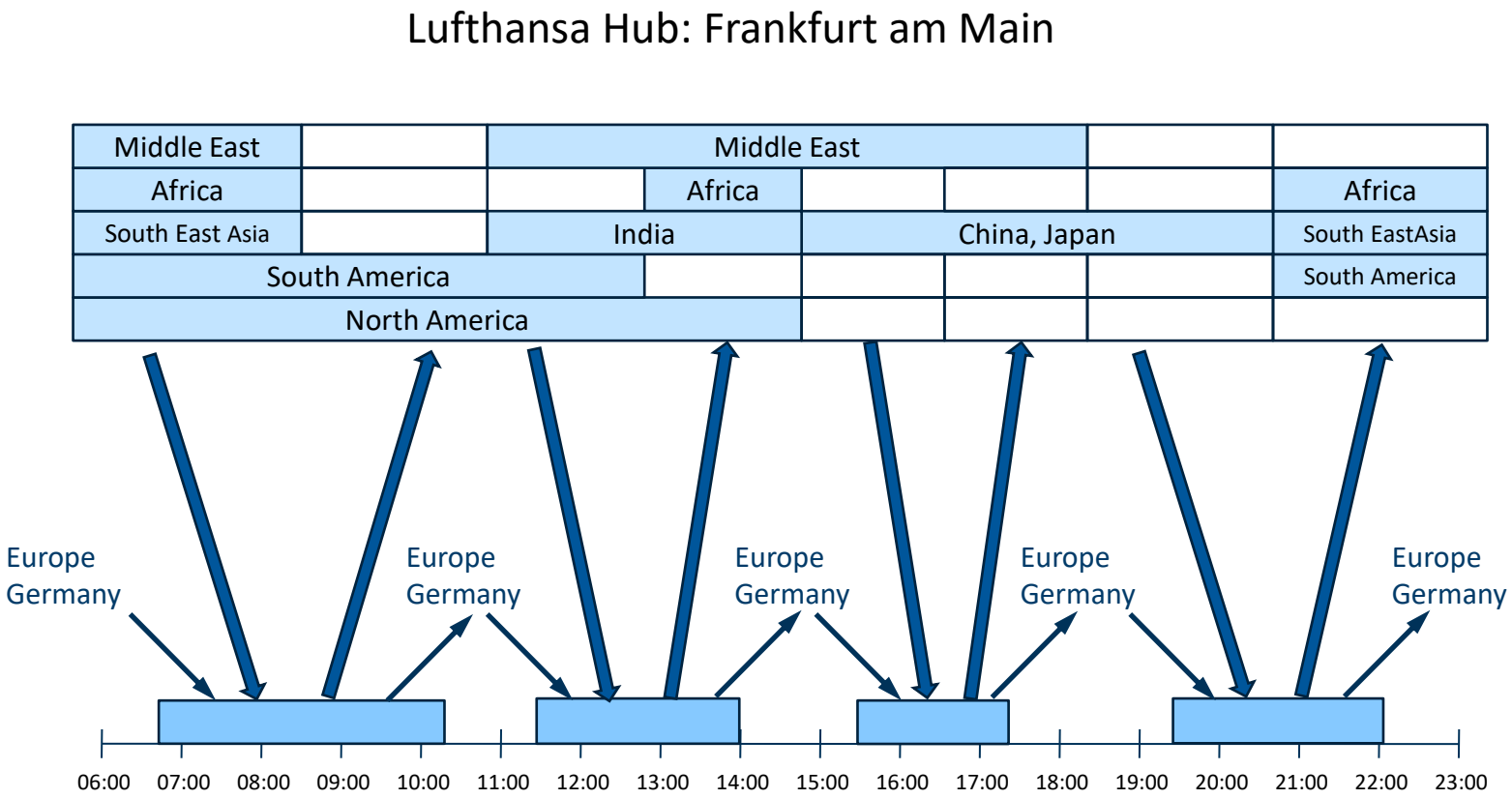
- Unattractiveness of connections to passengers
- High requirements of the performance of the infrastructure (Peak times)
- High complexity of the system
- Susceptibility to interference

Hub-&-Spoke VS. Point-to-Point

Aircraft manufacturers' philosophies



Hub Transportation at FRA (example)



Possible future business models

- many uncertain factors in future for aviation
- practical lecture air transprt scenarios aims to give a scope of possible developments

2018‘s course created a scenario (one out of many!) with two new business models:

Platform Airlines (PFA)

Concept: airline offers entire journey (single door-to-door ticket)

Ticket provider: airline

Revenue sources: aviation, ancillary, big part non-aviation

Airline share of travel chain:



Air Transport Provider (ATP)

Concept: internet companies offer door-to-door travel ticket, airline provides aircraft to internet companies, airline doesn‘t sell any tickets

Ticket provider: internet-company

Revenue sources: only aviation and ancillary

Airline share of travel chain:



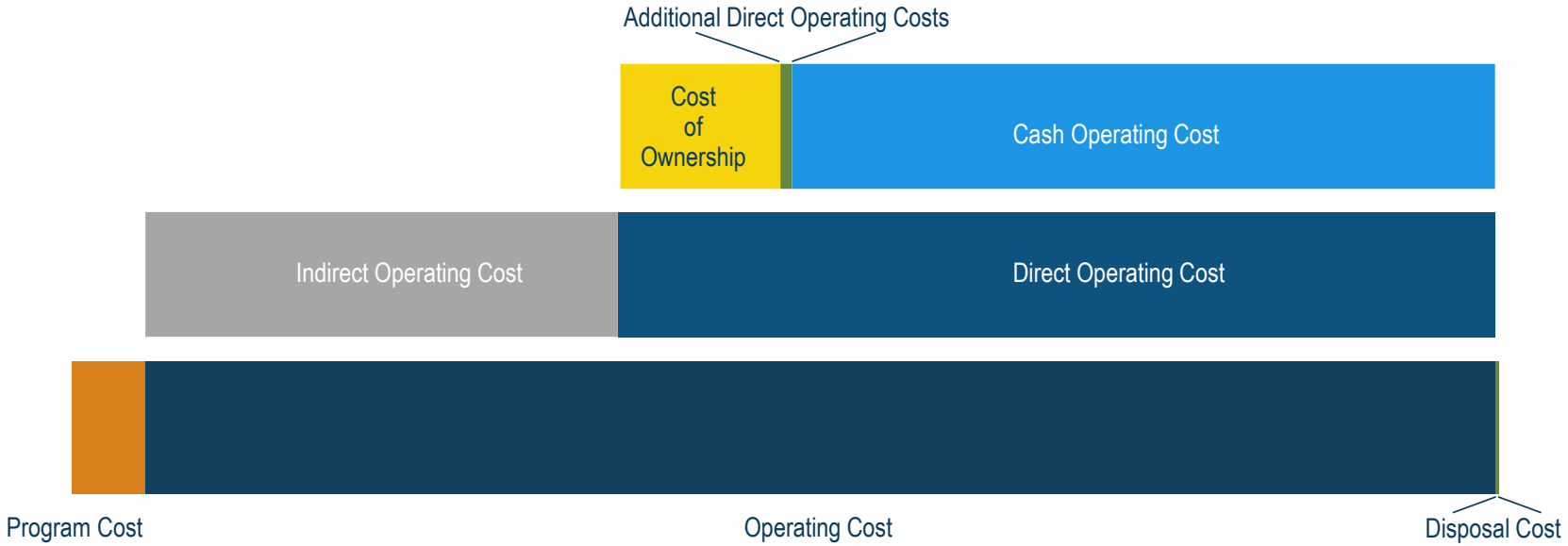
4. Commercial Aircraft Operators

4.2 Operating Costs



Life-Cycle Costs (LCC)

LCC distribution for a typical short-to-medium-haul transport aircraft*



- For civil transport aircraft, LCC are dominated by the Operating Costs
- Direct operating cost (DOC) have the highest share on Operating Costs
  - Share depending on operator and external factors i.e. fuel price
  - Decrease of Indirect Operating Costs (IOC) share can be observed (for AEA1996: 50%, 2006:40%)

* Underlying Assumptions:
616 NM stage length, operation: 20years, 35% share of IOC, fuel price 3.0 US$/gal, program costs covered by A/C market price, disposal cost equal to residual value (20 years)

Source: Bauhaus Luftfahrt

Operating Costs - Overview

Total Operating Costs (TOC)

Indirect Operating Costs (IOC)

Landed Properties & Ground Equipment

- Depreciation
- Maintenance

Passage / Freight

- Passenger, Baggage & Cargo Handling (Check-In)
- Booking, Sales
- Distribution
- Marketing

Administration & Overheads

- Buildings, Station Costs
- Training, Education
- Lease Costs

Direct Operating Costs (DOC)

Costs of Ownership (COO)

- Depreciation
- (Leasing)
- Interest Costs
- Insurance

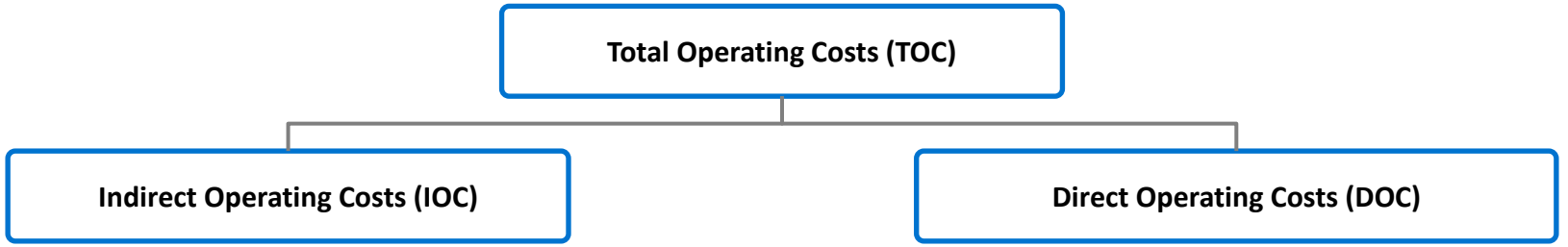
Cash Operating Costs (COC)

- Crew (Flight Deck, Cabin)
- Ground Handling
- Maintenance (Airframe, Engines)
- Navigation Fees
- Landing Fees
- Operating Materials (Fuel, Oil, etc.)

Additional DOC (ADOC)

- Local Noise Charges
- Local Emission Charges
- Emission Trading Charges

Operating Costs - Overview



Cost reduction is achievable through **corporate strategy optimization** with the aircraft operator.

Technological or efficiency improvements of aircraft do not contribute to cost reductions in this field.

Cost reduction is especially achievable through technological improvements of aircraft in the following areas:

- Low fuel burn
- Low maintenance efforts
- Safety
- Reliability
- Commonality

Flight Operating Costs of Selected Aircraft Types in U.S. (in 2017)

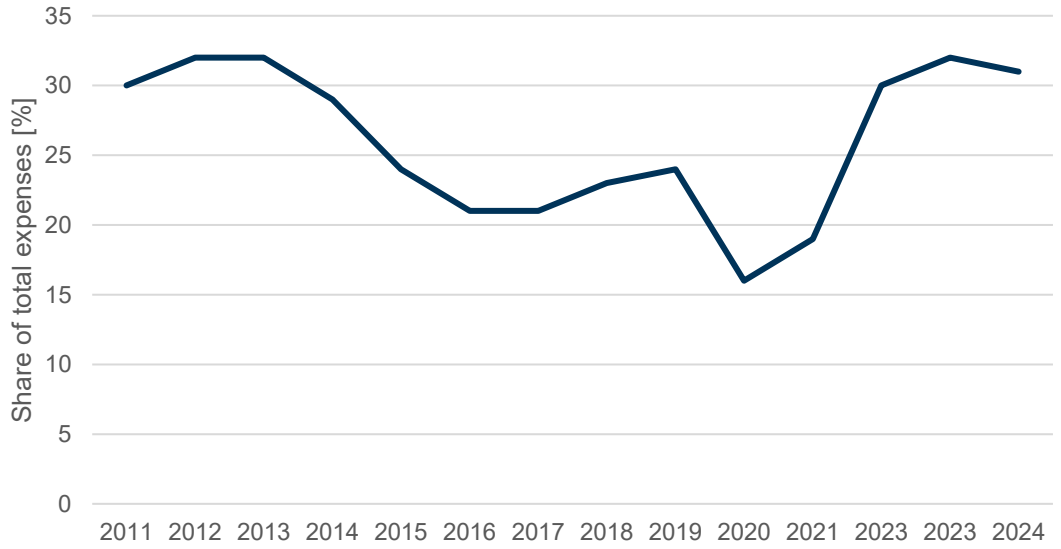
FOC = Flight Operating Costs, BH = Block Hours

	Aircraft Type	Seats	FOC per BH	FOC per Seat-Hour	Average Stage (mi.)	Utilization (BH / day)
	A319	134	3,435 $	25.63 $	885	9.3
	737-700	141	3,561 $	25.26 $	721	10.5
	757-200	178	4,655 $	26.15 $	1,767	10.3
	A330-200	263	7,072 $	26.89 $	3,465	12.6
	787-9	261	7,369 $	28.23 $	5,729	13.4
	747-400	375	12,179 $	32.48 $	5,723	10.7

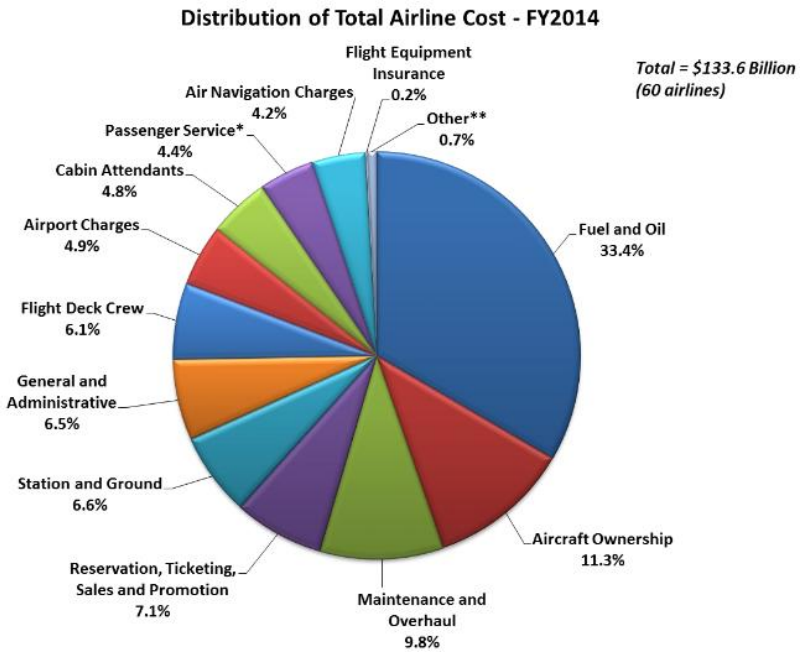
Source: The Airline Monitor Commercial Aircraft Data Base, aeroinside.com, planespotters.net

Development of Aircraft Operating Costs

Share of fuel costs on operating costs



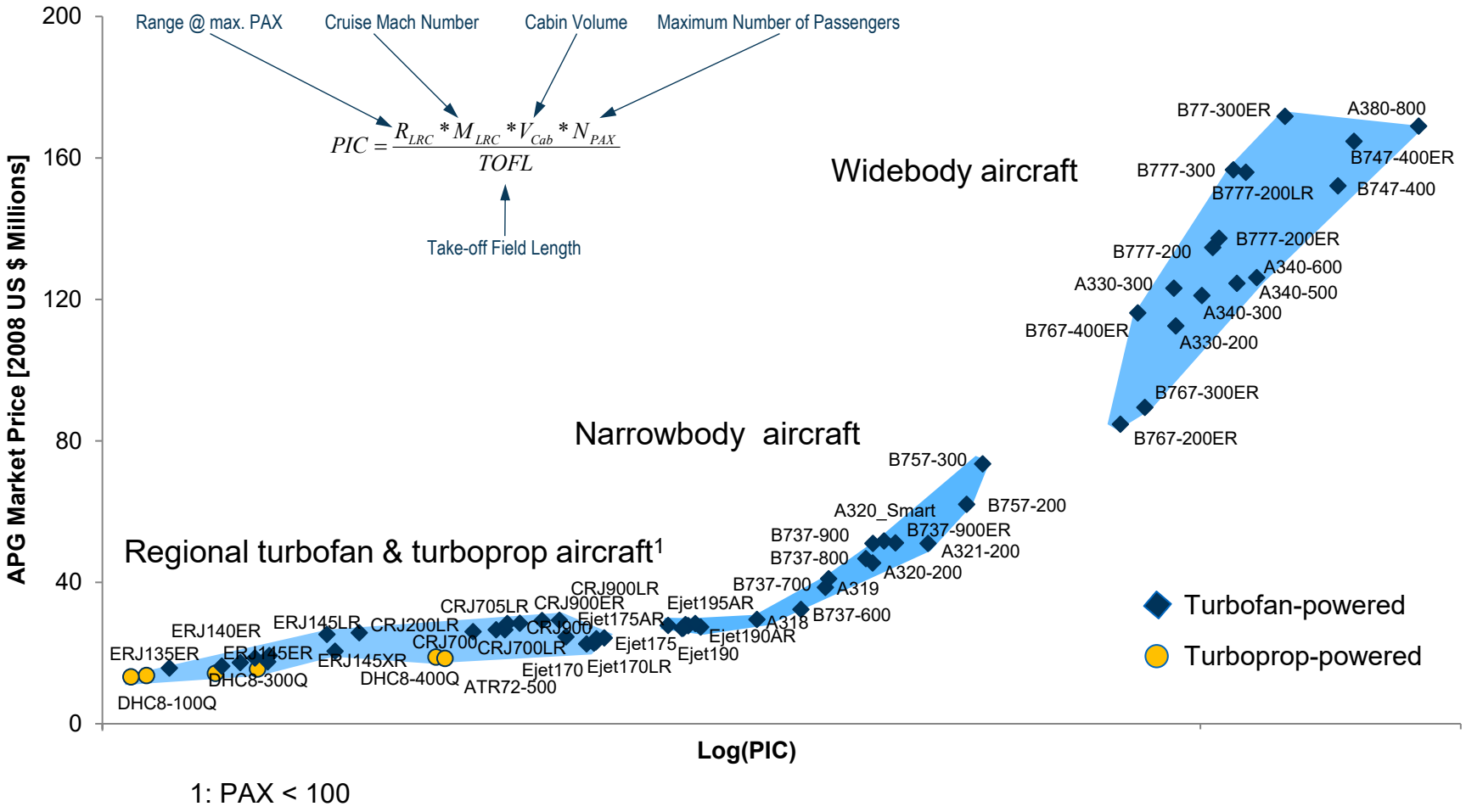
Distribution of Total Airline Costs 2014



Fuel costs have been highly volatile within the recent years.

Source: The Airline Monitor Commercial Aircraft Data Base, IATA

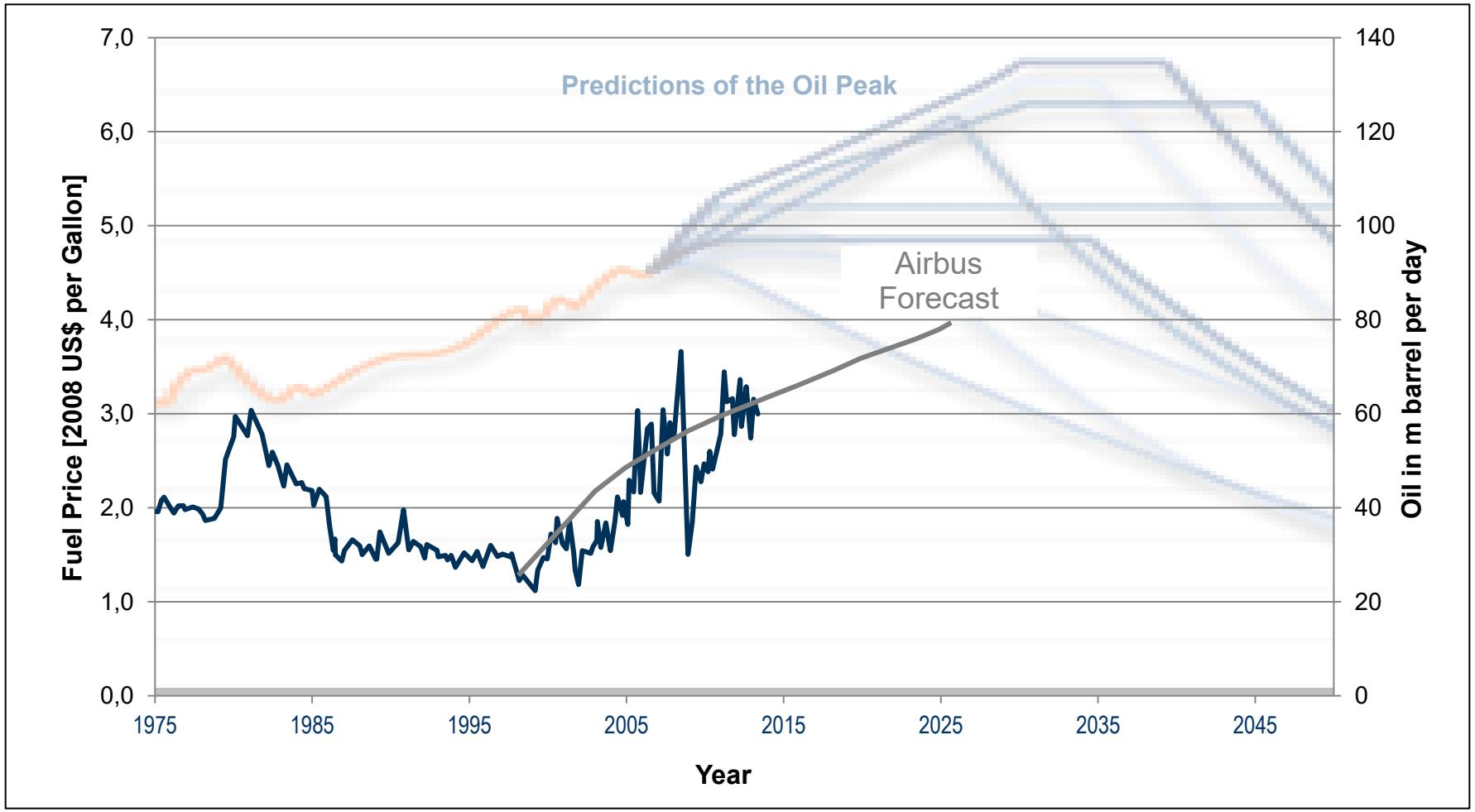
Aircraft Market Price VS. PIC



1: PAX < 100

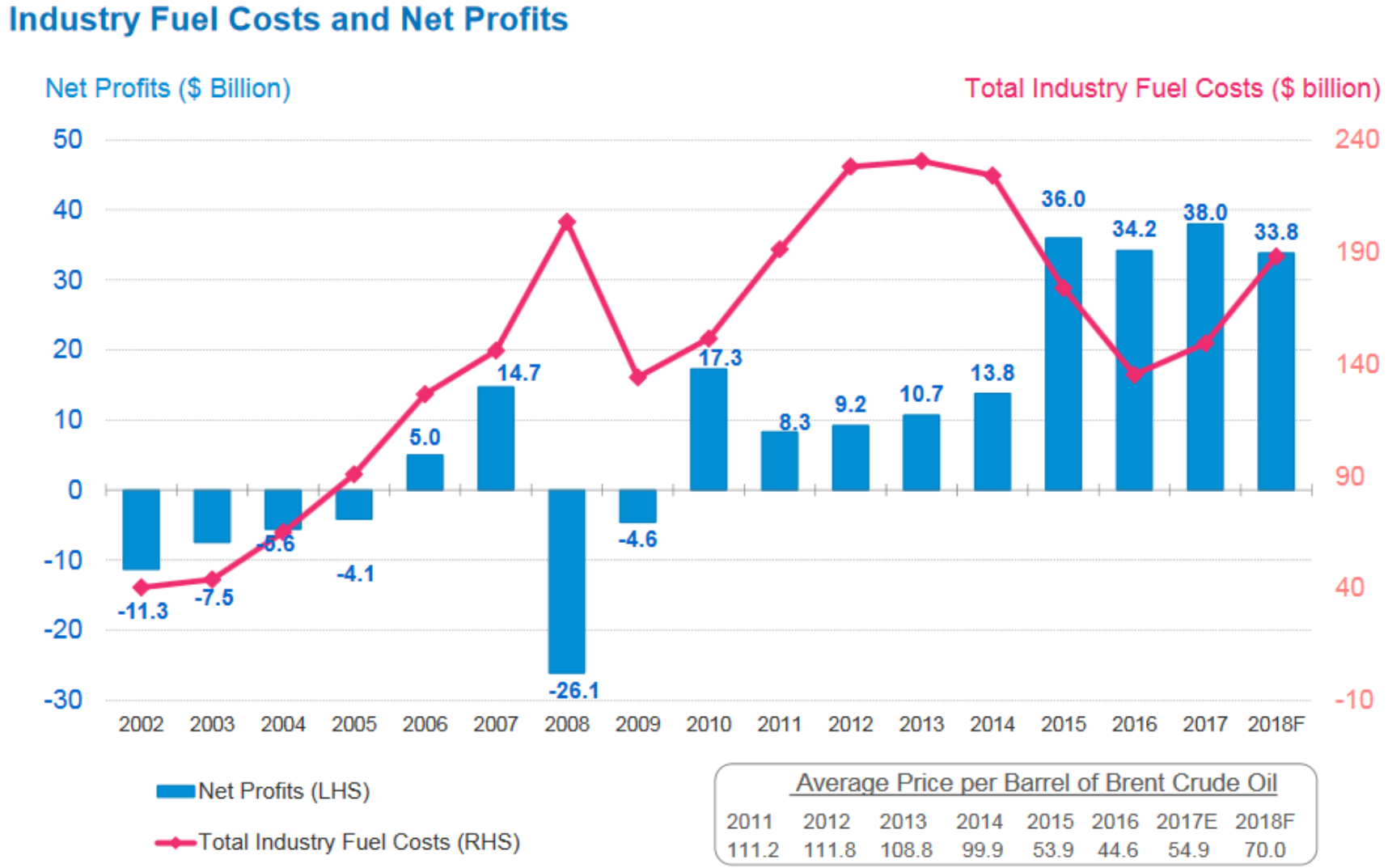
Source: Isikveren A.T. et al. 2003: Productivity Metrics for Business and Regional Aircraft, SAE Paper 2003-01-3063

Development of Fuel Price



Data based on: U.S. Department of Energy; U.S Bureau of Labor Statistics

Fuel Costs



Average Price per Barrel of Brent Crude Oil

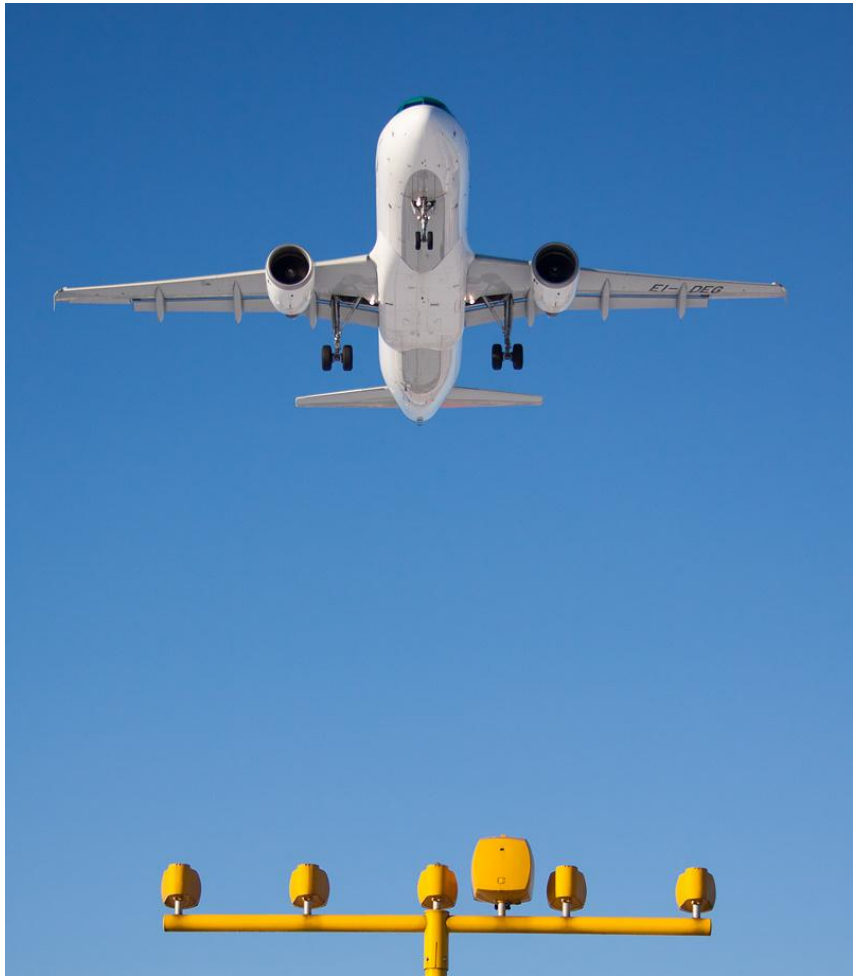
2011	2012	2013	2014	2015	2016	2017E	2018F
111.2	111.8	108.8	99.9	53.9	44.6	54.9	70.0

Source: IATA FACT Sheet: Fuel, June 2018

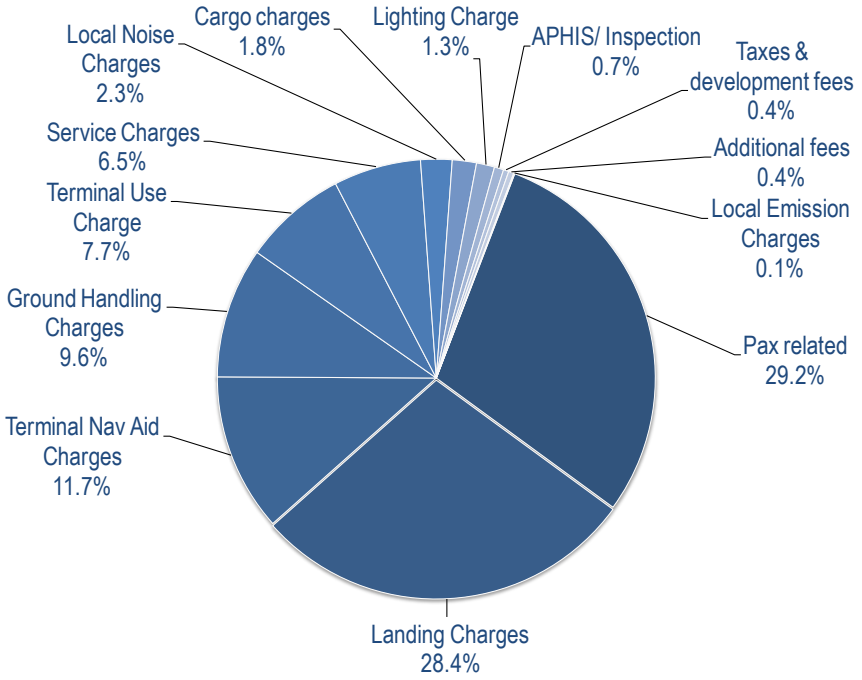
Airport Charges



Mean Airport Charges for an Airbus A320



Source: Bauhaus Luftfahrt, OAG database

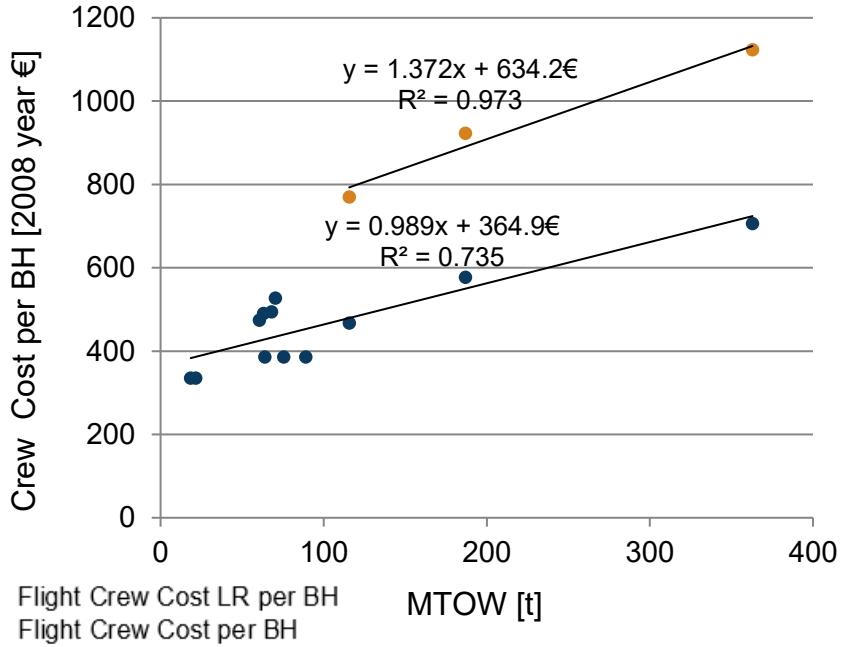


Strong increase in environmental (noise and emission) charges over the last 40 years.

Crew Costs (Flight Deck & Cabin)

Variation of crew cost depending on

- Airline Business Model
- Market Sector
- Airline Age
- Geographical region (crew´s base country)
- Aircraft type



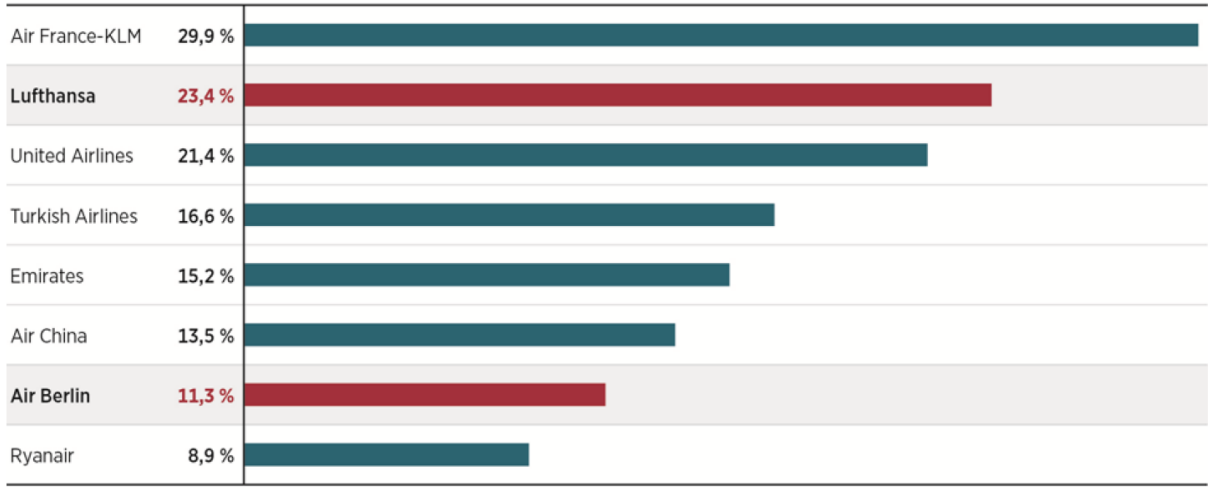
Number of Crew

- 20 seats and more: 1 cabin crew member
- Each 50 seats (per deck): 1 additional cabin crew member (Canada: per 40 PAX onboard)
- Flight crew:
  - 8h to 12h: 3 (typically 1 Cpt, 2 F/Os)
  - >12h: 2 Cpts and 2 F/Os

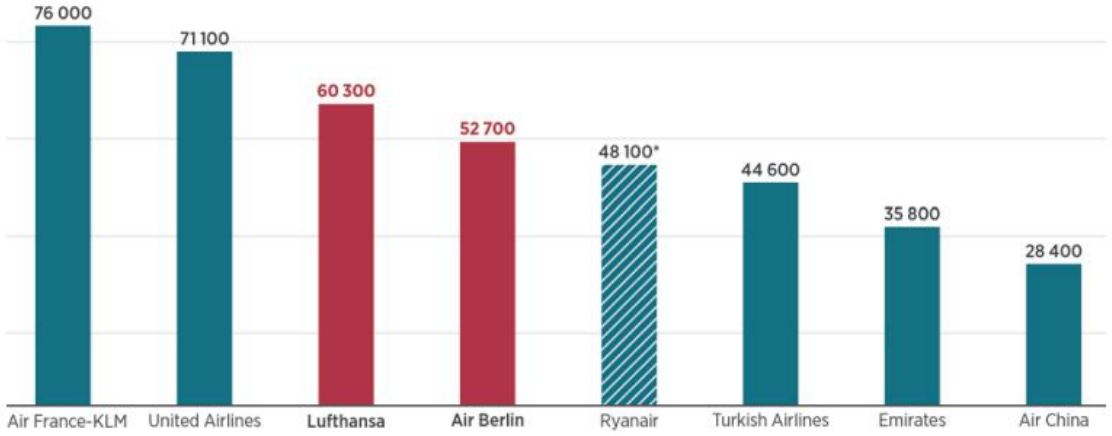
Picture Reference: Transport Studies Group, 2008. Innovative Cooperative Actions of R&D in EUROCONTROL Programme CARE INO III - Dynamic Cost Indexing. Aircraft Crewing - Marginal Delay Costs, London – adapted by Wesseler (2011)

Crew Costs: International Comparison

Share of Crew Costs in Total Turnover



Crew Costs per Employee [€]

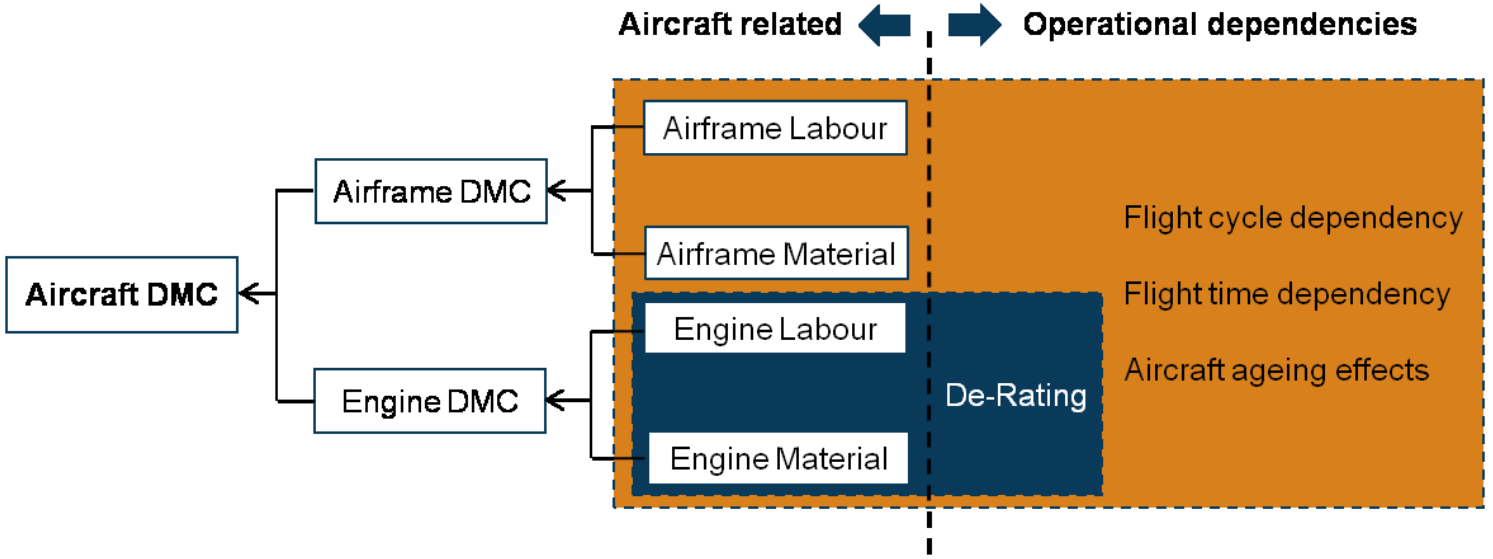


Source: Handelsblatt Research Institute (2014)

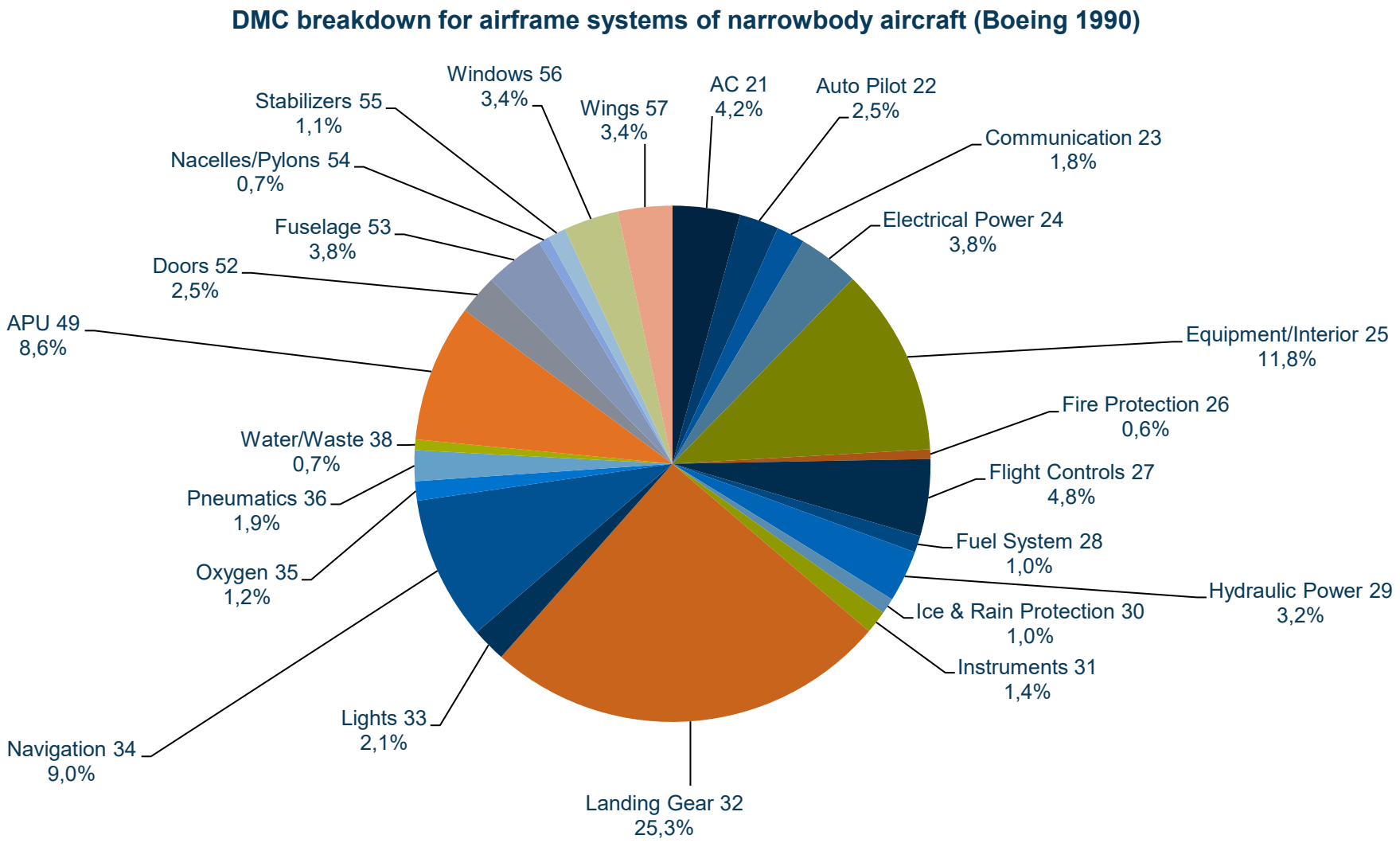
Maintenance Costs

Expenditures for maintenance and overhaul

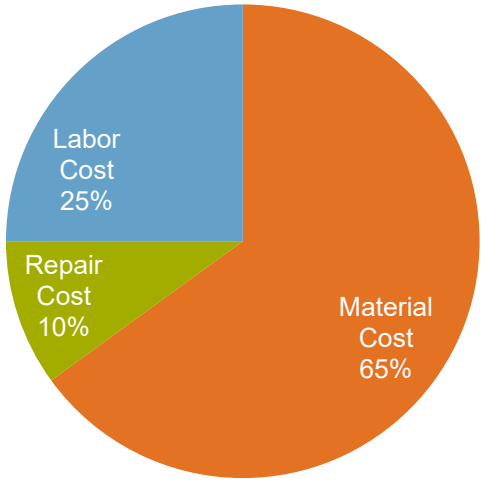
- **Direct Maintenance Cost (DMC)**
  - Compromises labor and material cost associated with airframe and engine
- **Indirect Maintenance Cost (IMC)**
  - Overhead, administration, tooling, testing equipment, quality controls
- **Aircraft Down-time** (associated costs not included)



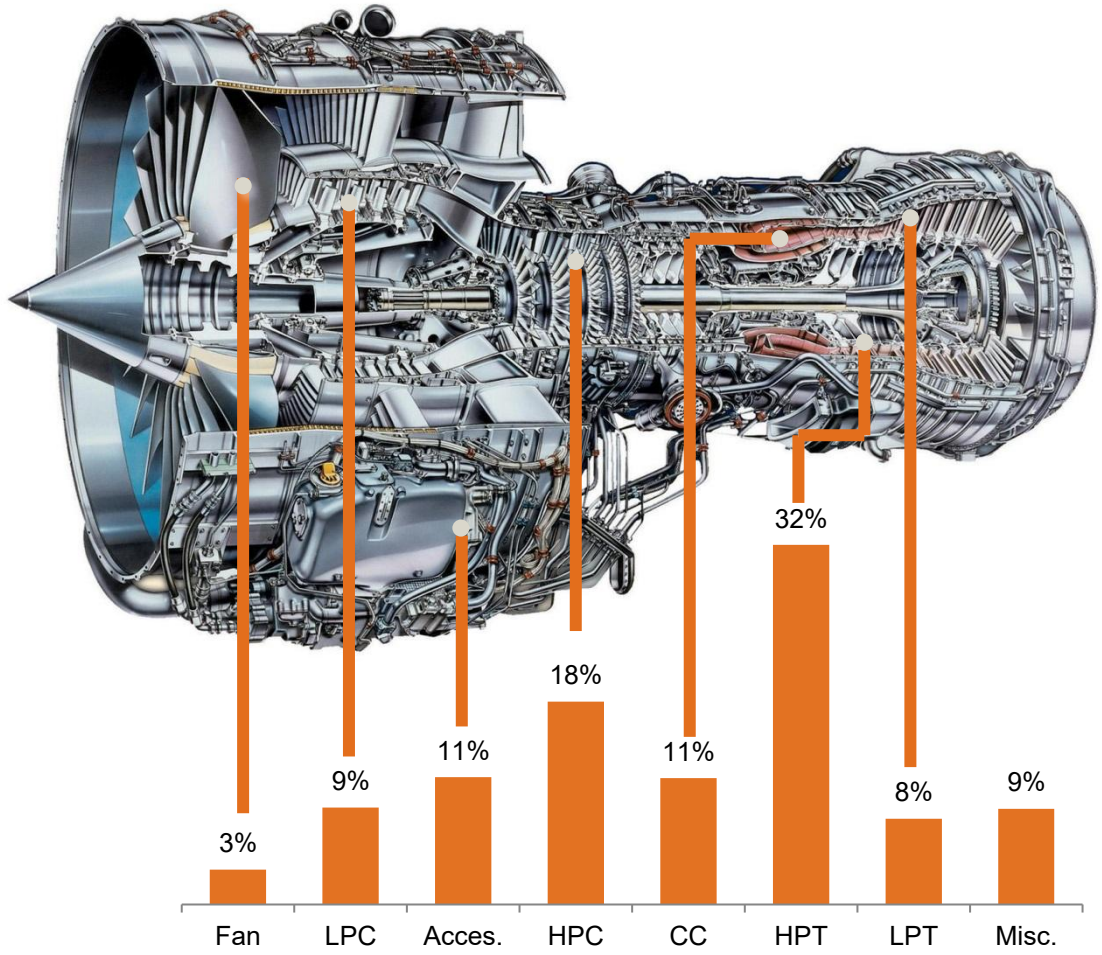
DMC Cost Share



Engine Maintenance Costs



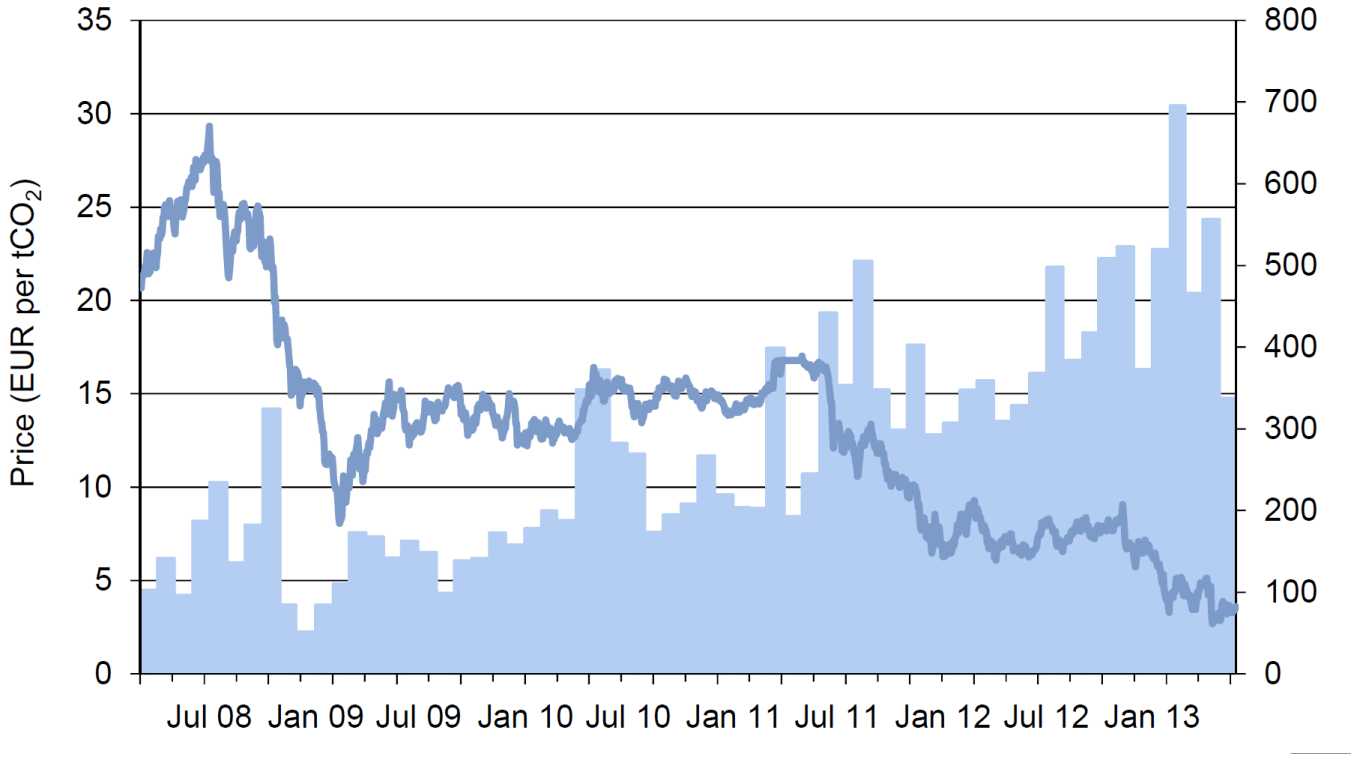
Engine DMC mean values for repair, material, and labor cost (Ackert, 2011)



Cost share of IAE V2500-A5 engine (MTU Maintenance 2013)

ADOC: Emission Trading Scheme

- Additional costs for emitting CO_2
- Aviation included in EU ETS since 2012, but only for Domestic EU Flights
- Allocation of free allowances to European Airlines (85% in 2012, 83% in 2013-2020)



Example of Cost Analysis

Airbus A380-800 vs. Boeing 747-8

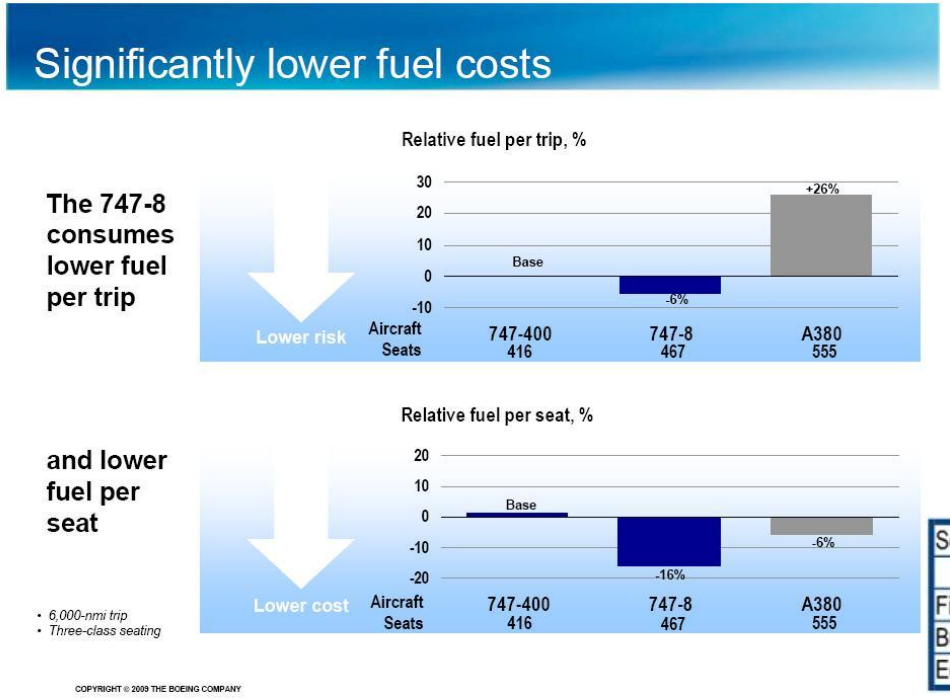
Cost-related figures shown in

- US$ per trip
- US$ per block hour (BH)
- US$ per seat/PAX
- US$ per nautical mile/seat-kilometer

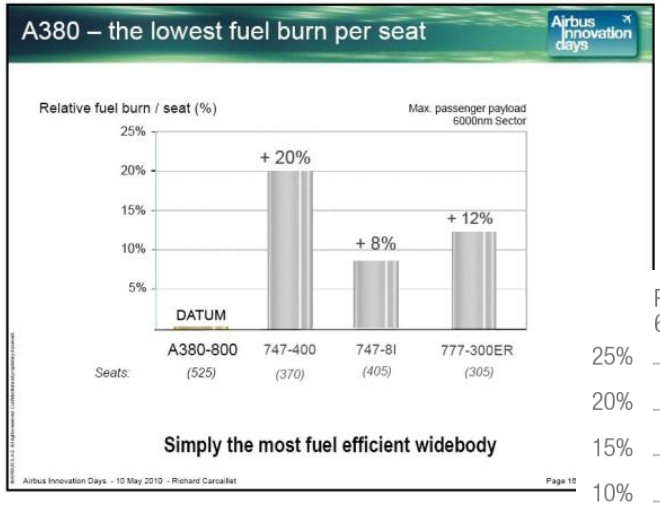


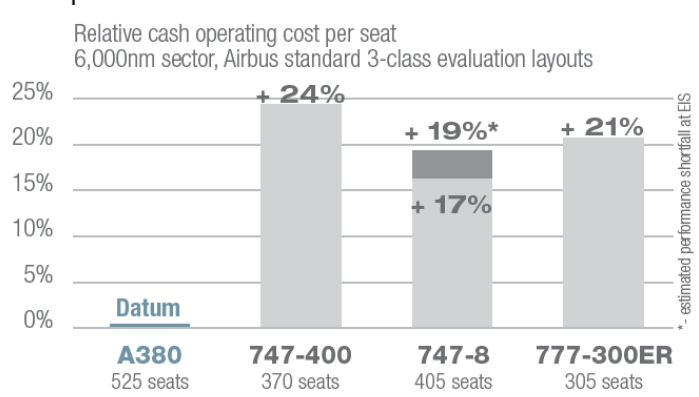
Airbus A380-800 vs. Boeing 747-8

Boeing View



Airbus View





Seat Pitch Assumptions for Economics Inches		
	Boeing	Airbus
First Class	61	81
Business Class	39	61
Economy	32	32

…considers only fuel; lower trip fuel while 747-8 has 24.7% lower MTOW compared to A380	…considers only cash operating costs, no cost of ownership
Consistent calculation of seating numbers but 61” in FC?	…cost figures per seat, much lower seat numbers for Boeing aircraft

…and what about cargo revenues, fuel prices, and other charges?

4. Commercial Aircraft Operators

4.3 Airline Planning and Strategy



Airline Planning Tasks

The most important planning decisions faced by airline managers can be categorized as follows:

- ***Fleet Planning***: What type of aircraft to acquire, when and how many of each?



- ***Route Planning***: Where to fly the aircraft profitably, subject to fleet availability constraints?



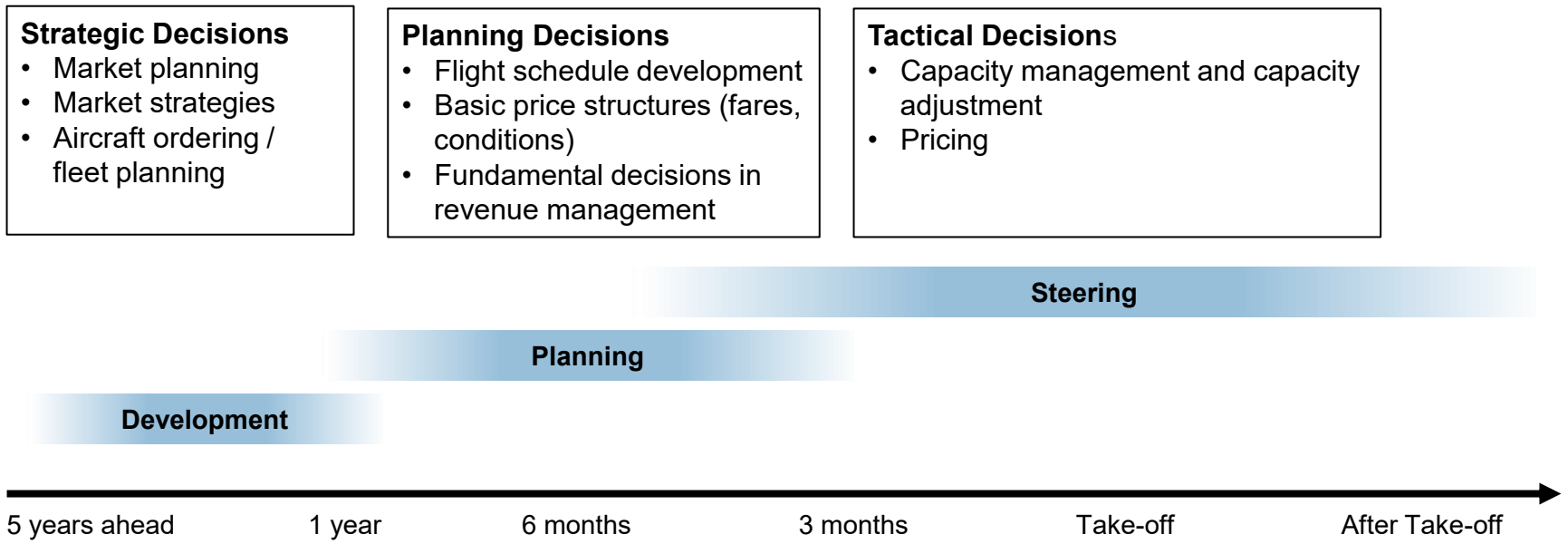
- ***Schedule Development***: How frequently and at what times on each route should flights be operated, subject to operational and aircraft limitations?



Source: Belobaba (2009)

Stages of Airline Planning

Airline Planning is made on long term. It starts with **long-term decisions of strategic nature** and continues with **planning and steering activities** of the airline management. At the end of the airline planning process, detailed knowledge about the **airline's fleet composition** as well as the **network** to be served and the **flight schedule** to be offered is available.



Airline Planning: Influence Factors

Market / Demand
- Strategy / Market Share
- Amount of Passengers
- Yield per Passenger
- Demand Behavior
- Competitors
- Cooperation

Operations
- Aircraft Performance
- Crew
- Technical Requirements
- Dispatch
- Cost Structures

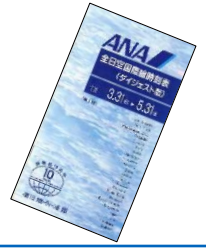
Restrictions
- Fleet
- Slots
- Block Times
- Ground Times
- Airport Infrastructure
- Curfews
- Traffic Rights

Quality
- Punctuality
- Passenger Connections
- Baggage Connections

- Aircraft / Fleet
- Frequencies
- Departure Time
- Hub Structure



Timetable of Scheduled Flights



Airline Planning Tasks

- **Fleet Planning**
- Route Planning
- Schedule Development

Questions in Airline Fleet Planning

Please buy one or more aircraft units with a budget of 145 Mio. US$!

Would you buy one of this type?



Or would you buy several ones of this type?



Do you pay cash or by instalments?

Do you pay in Euros or Dollars?

What are your decision criteria?

Key Questions

An airline has to treat the following questions in fleet planning:

- **When** should it buy an aircraft?
- **How many aircraft** should it buy?
- **Which types of aircraft** should it buy?
- Which type of aircraft should it **use on which route**?



Cost Factors of an Aircraft Fleet

Operating Costs	• Fuel, crew, maintenance, airport charges, take-off and landing fees, etc.
Reserves (Back-Up)	• Size of reserves, provision of spare aircraft in order to ensure stable traffic operations
Flexibility of Equipment Replacements	• Expenses and effort necessary to replace defective aircraft (crew licensing, capacity, range, etc.)
Maintenance	• Maintenance intervals, capacity utilization of maintenance operation, experience concerning specific aircraft types
Degree of Standardization	• Crew training, cooperation with manufacturers, technical inspections

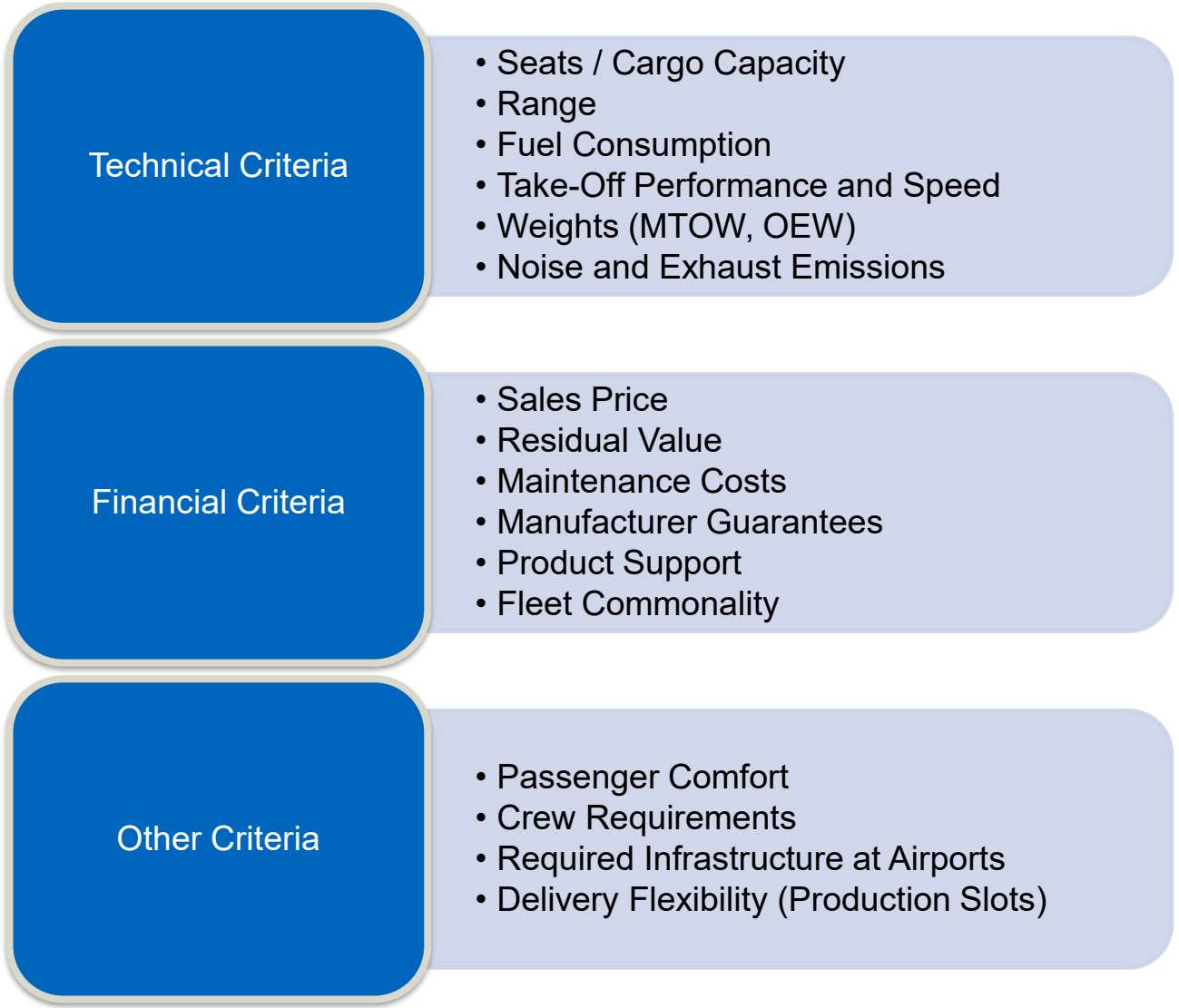
Top-Level Goals of Fleet Development…

… from airline perspective:

- Minimum Operating Costs
- Modern aircraft
- Economies of scale in acquisition
- Fleet commonality
- Capacity and range flexibility
- Minimum capital investment
- Risk management
- Product differentiation

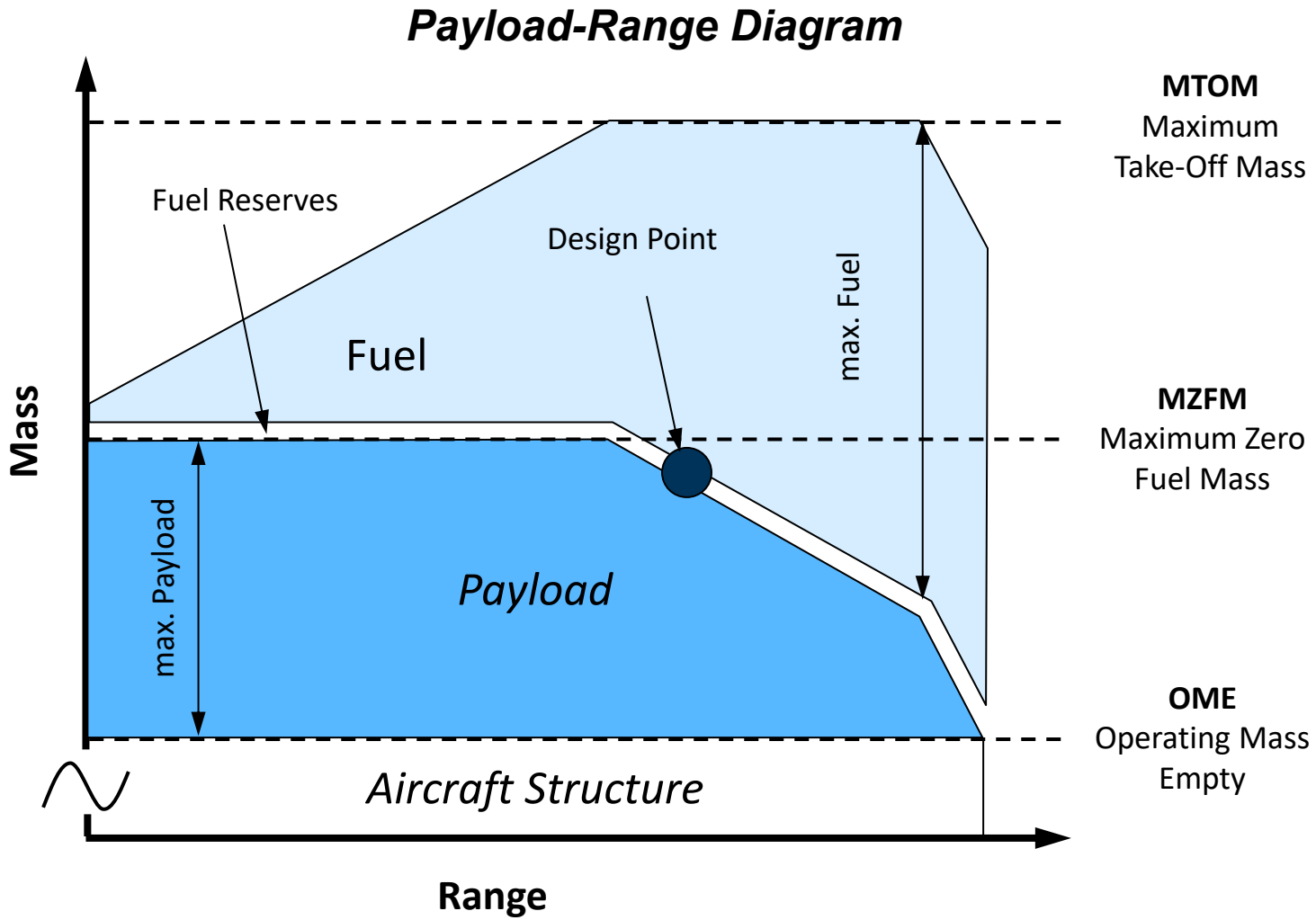


Aircraft Selection Criteria (1)



Aircraft Selection Criteria (2) (see Lecture 3, Mission Analysis)

Two major selection criteria: **Payload Capacity** and **Range Capability**



Aircraft Selection Criteria (3)

The most important criterion is the **economic profitability** of an aircraft that is a direct result of its **Direct Operating Costs**.

The airline has to do an economic and financial trade-off before buying a new aircraft:



Higher Ownership Costs

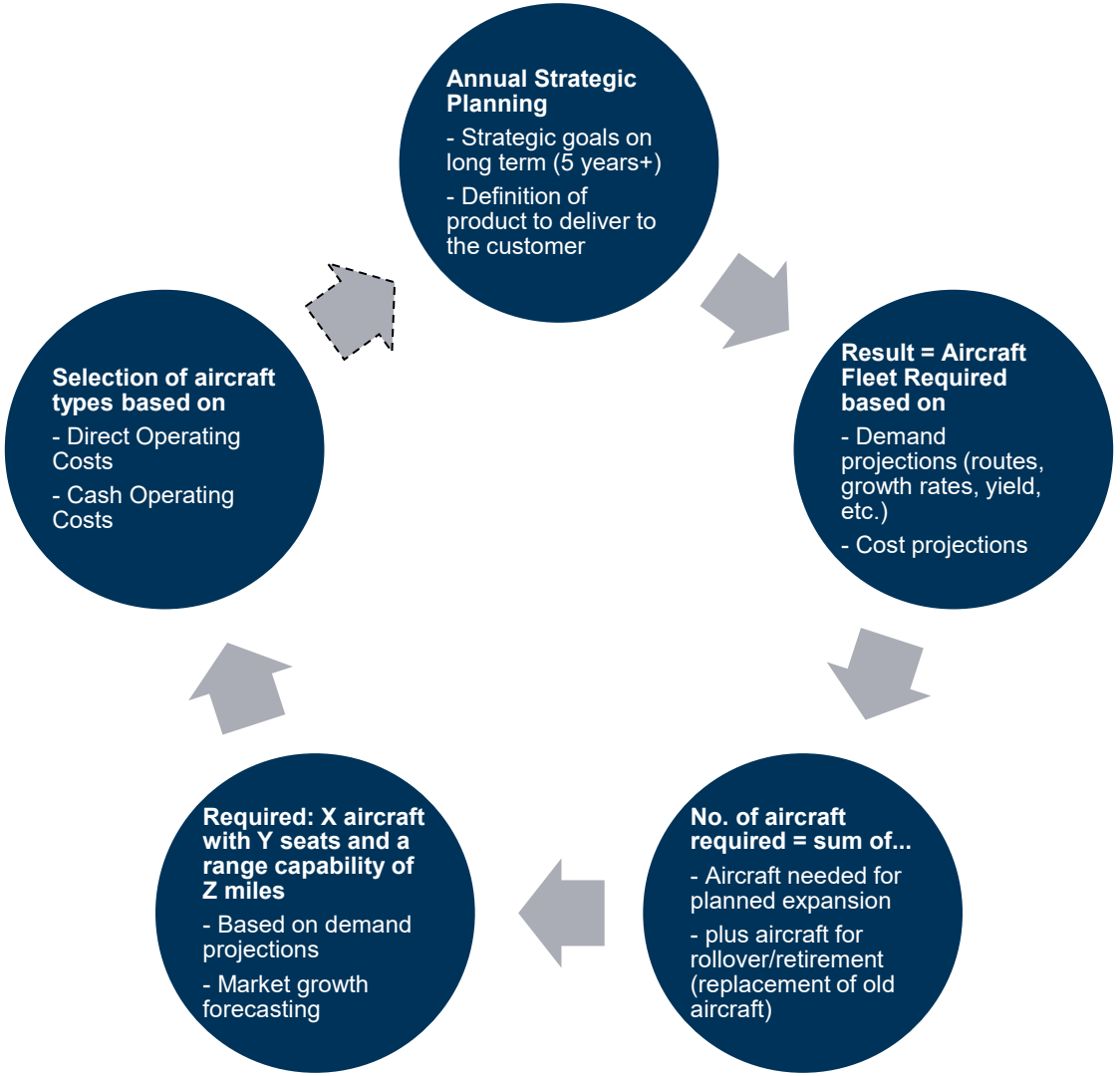
Lower Operating Costs

Aircraft Selection Criteria (4)

Further criteria also influence the airline's selection of aircraft:

- **Noise and emissions performance** of the aircraft
- **Cabin seat configuration**, e.g., 6-abreast vs. 5-abreast (A320 vs. Jumbolino)
- **Propeller vs. Jet**, e.g., ATR vs. Bombardier Regional Jet
- **Engine position**, e.g., Boeing 737 vs. Boeing 717
- **Variability** of utility elements (e.g., kitchen)
- **"Passenger appeal"** (*Have you already flown with an Airbus A380??)*
- Influence on fleet selection by **politics**
- …

The Airline Fleet Planning Cycle



Critical Inputs required for Fleet Planning

The most critical inputs include

- **the air traffic forecast**
- **yield forecast**
- **estimated aircraft productivity**
- **operating costs**

These data usually come from databases managed by different departments within the airline, which can make the planning process difficult to implement in practice.

Source: Belobaba (2009), Clark (2007)

Two fundamental Approaches to Fleet Planning

1. The **‘top-down’ or ‘macro’ approach** based on relatively high-level aggregate analysis

2. The **‘bottom-up’ or ‘micro’ approach** based on much more detailed analysis of data and forecasts by flight and route

see Lecture 2, Market Outlook Methodologies

Source: Belobaba (2009), Clark (2007)

Phases of Aircraft Acquisition



Aircraft Specification	Development and Production Inspection	Technical Acceptance, Transfer, Retro Fitting	Launch, Gaining Experience, Technical Improvements
• Technical part of the sales agreement with all technical details of the aircraft concept • Description of “Buyer-furnished Equipment” (e.g., seats)	• Airline representatives are positioned directly with the aircraft manufacturer • Cooperation between the manufacturer and the airline (e.g., handling technical difficulties) • Development progress monitoring and participation in aircraft testing	• Examination of production defects • Installation of “Buyer-furnished Equipment” (seats, galleys, in-flight entertainment system, etc.)	• Keeping a technical log book • Error and failure reports • Deduction of technical instructions and improvements • Appraisal and application of the manufacturer guarantee

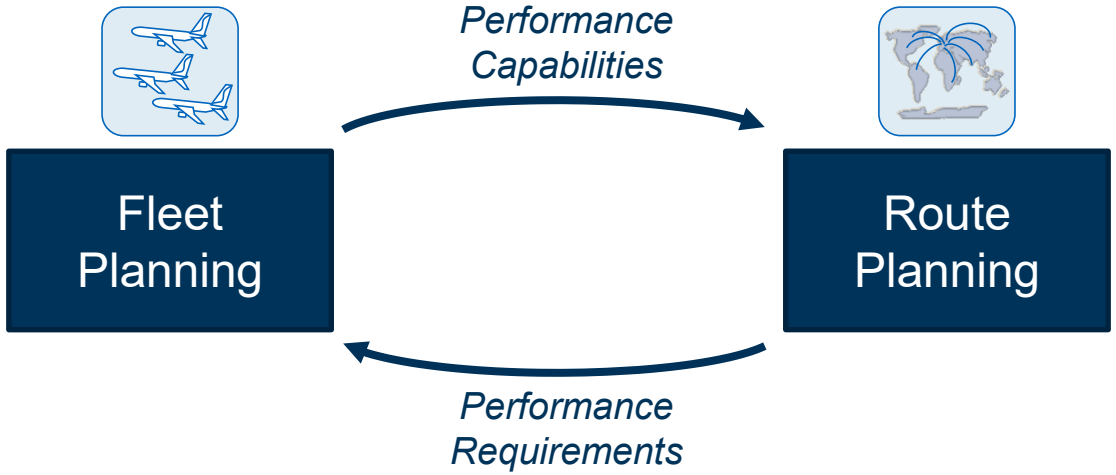
Airline Planning Tasks

- Fleet Planning
- **Route Planning**
- Schedule Development

Route Planning

Goal of route planning: Airline defines the route network to be flown.

- Aircraft & fleet performance characteristics for specific routes provide a **feedback loop** to fleet planning decisions!



- Route planning is made both on **long-term** (*strategic planning,* time horizon: several years) and **short-term** (*tactical planning*, time horizon: weeks to months) basis.

Source: Belobaba (2009), Clark (2007)

Route Evaluation (1)

Economic considerations dominate route evaluation

> Does the route own the potential of being profitable when served?

Important input data required for route evaluation:

- Passenger and cargo demand forecasts
- Expected market share
- Expected revenues
- Competitor's behavior

It is important **not to consider the O-D passengers only**, but also the **connecting passengers** (hub-&-spoke network).

> There might be routes that are not profitable by themselves (considering only O-D passengers), but that make an important contribution as "feeder operations" to other routes within the airline network.

Source: Belobaba (2009), Clark (2007)

Route Evaluation (2)

Criteria for route evaluation:

- **Incremental profitability** (on short term): estimating the opportunity cost of taking aircraft from one route to another.
- **Technical capability and availability of aircraft**: adequate range and payload capacity.
- **Operating cost characteristics of aircraft**: estimating the operating costs when operating the aircraft on a specific route.
- **New route?** Consider additional costs of establishing airport facilities, sales offices, and staff relocation.
- **Political and legal restrictions, bilateral agreements, limited airport slots, etc.**
- **Strategic considerations**: evaluate the competitor's behavior and decide how to (re-)act

Source: Belobaba (2009), Clark (2007)

Route Profitability Model

…used by airlines to select routes to **maximize the total profits**,

given a set of **candidate routes** and **estimated demands**,

subject to **fleet and capacity constraints**.

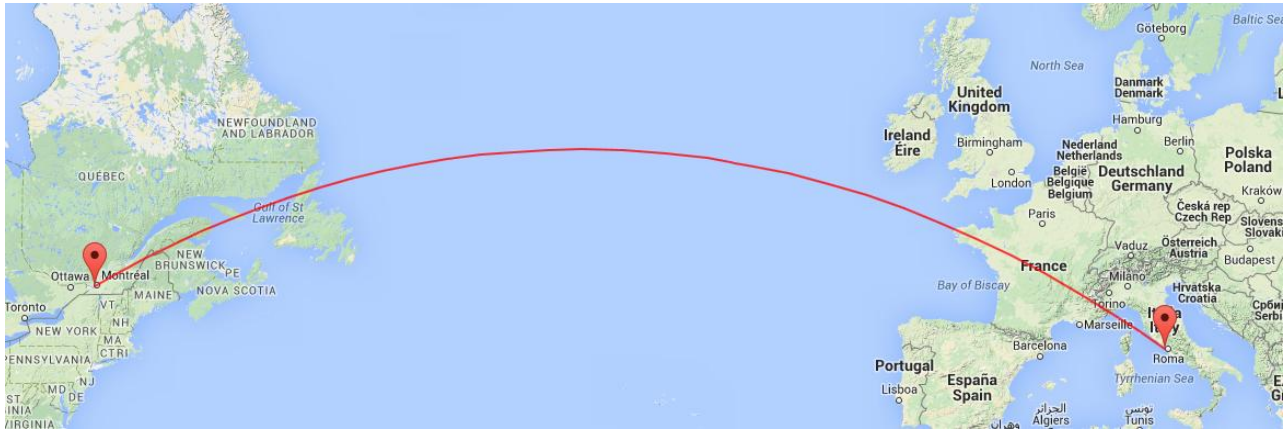
Results depend entirely on the quality of the input data. *Garbage in, garbage out!*

> Need to do sensitivity analyses!!

Example: Route Profitability Analysis

Route Proposal: Montreal (YUL) to Rome (FCO), non-stop

6,600 km of stage length



Source: Belobaba (2009), Clark (2007), Google Maps

YUL – FCO: Revenue Calculations

Demand and fare estimates for year	Annual demand	Average revenue (one way)	TOTAL REVENUE
Total YUL-FCO local O-D passengers (both directions)	102,000		
Expected market share for one daily flight	70%		
Local YUL-FCO passengers on new flight	71,400	$450	$32,130,000
Additional Traffic			
Connections North American cities behind YUL to/from FCO	24,000	$425	$10,200,000
Connections to/from YUL beyond FCO	12,000	$400	$4,800,000
Connections behind YUL to/from destinations beyond FCO	4,500	$375	$1,687,500
Total passengers (both directions)	111,900		$48,817,500
Additional cargo revenue (estimated 10% of Pax revenue)			$4,881,750
		TOTAL REVENUES	**$53,699,250**

Source: Belobaba (2009)

YUL – FCO: Operating Cost and Profit Calculations

Inputs and assumptions

Aircraft type	**Boeing 767-300ER**
No. of seats	210
Total annual flights (each direction) (reflects 98% completion of daily schedule)	358
Block Hours YUL - FCO	08:00
Block Hours FCO - YUL	09:00
Stage length YUL / FCO	4,086

Aircraft operating costs per Block Hour

Crew costs	$890
Fuel / oil	$3,280
Ownership	$870
Maintenance	$710
Total per Block Hour	**$5,750**

Indirect operating costs

Passenger service	$0.015	per Revenue Pax Mile (RPM)
Traffic servicing	$22	per enplanement
Aircraft servicing	$1.800	per departure
Promotion and sales	9.00%	of passenger revenues
General and administrative	$0.002	per Available Seat Mile (ASM)



Calculated measures (per annum)

Annual flights	716
Block Hours	6,086
RPMs	457,223,400
Pax yield	0.1068
ASMs	614,370,960
Seat departures	150,360
Passengers enplaned	111,900
Average load factor	74.42%

TOTALS

DIRECT OPERATING COSTS	$34,994,500
PAX SERVICE	$6,858,351
TRAFFIC SERVICE	$2,461,800
AIRCRAFT SERVICE	$1,288,800
PROMOTION / SALES	$4,393,575
GEN ADMINISTRATION	$1,228,742
Operating costs	**$51,225,768**
Operating profit	**$2,473,482**
Operating margin	**4.8%**

Source: Belobaba (2009)

Airline Planning Tasks

- Fleet Planning
- Route Planning
- **Schedule Development**

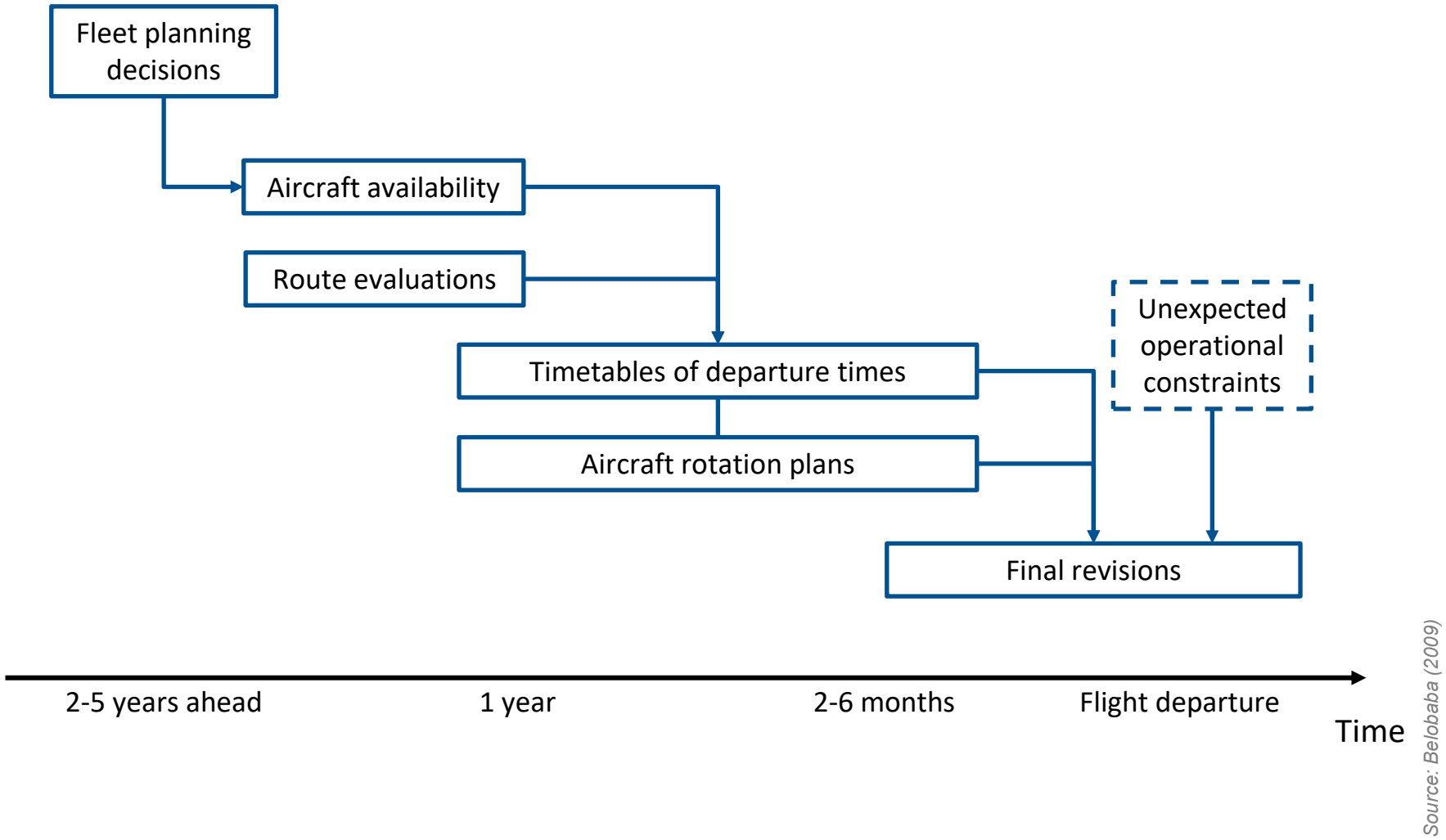
Schedule Development (also see Lecture 2: A/C Utilization)

Four major tasks:

- ***Frequency planning***: how often should the airline operate flights on the selected route network?
- ***Timetable development***: at what times should flight departures be scheduled?
- ***Fleet assignment***: what type of aircraft should be used for each departure time?
- ***Aircraft rotation planning***: how should each aircraft type be flown over the airline's network in order to ensure an overall balance of aircraft arrivals and departures et each airport?

Source: Belobaba (2009)

Schedule Development: Chronological Overview



Source: Belobaba (2009)

Frequency Planning

Based on the outputs of the **route evaluation process**, the airline should already have estimates of the **total "local" demand** between the origin and destination served by the route under consideration.

The airline's supply decision consists of two choices: **number of departures per day** and **number of seats** to be offered on each departure





Higher frequency / lower seat capacity
=
Increasing Pax convenience
&
Increasing operating costs



Lower frequency / higher seat capacity
=
Decreasing Pax convenience
&
Decreasing operating costs

- **Higher ticket prices: business travelers**
- **Higher market share & load factor**

- **Lower ticket prices: leisure travelers**
- **Lower market & load factor**

Source: Belobaba (2009)

Timetable Development

Initial goal of the airline: provide departures at **peak periods** (usually at around 09:00 am and 5:00 pm), especially for business travelers

This is not always possible: slot restrictions, limited amount of aircraft in fleet, etc.

- Trade-off required:



Source: Belobaba (2009)

Important issues to consider:

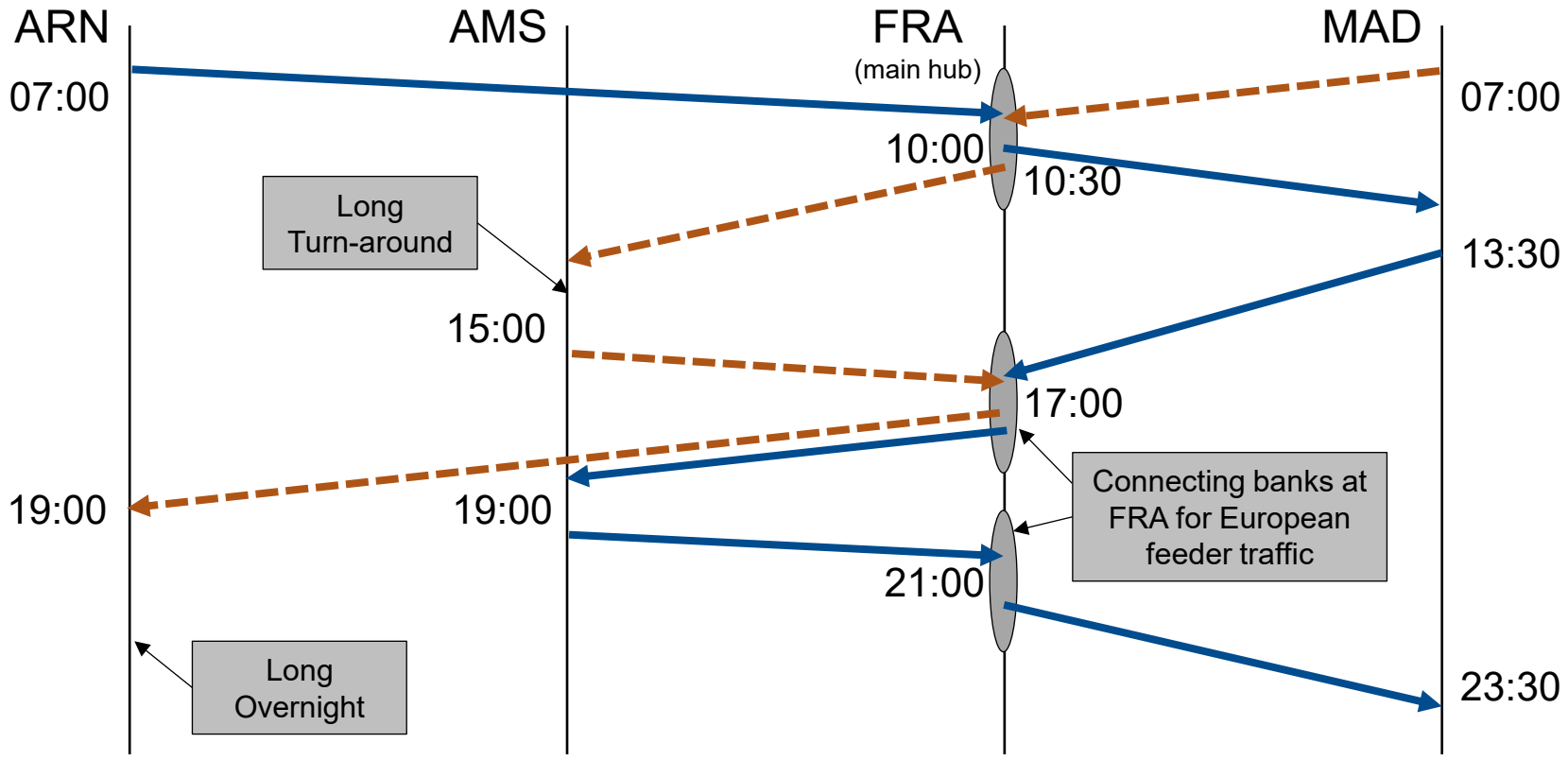
- Turn-around time at the airport (de-/boarding, cleaning, refueling, etc.)
- Schedule quality for return flight (resulting in a reduced load factor)
- Connecting times (hub network)
- Time zone differences between origin and destination airport
- Regulatory constraints (e.g., night curfews at certain airports)
- Crew scheduling requirements & aircraft routine maintenance

No computer tool available that could capture the full complexity of the problem!

Timetable Development: The Schedule Map

Schedule maps are a visual representation of flight legs that incorporate aircraft movement in terms of both geography and time.

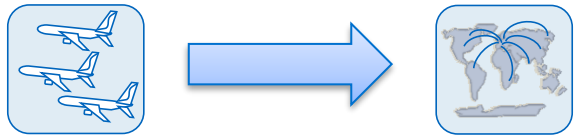
EXAMPLE: two aircraft operating as feeders in a small European network



Source: Belobaba (2009)

Fleet Assignment and Aircraft Rotations

Fleet Assignment Problem (FAP): determines **the type of aircraft** to be flown on **each flight leg departure**, given a planned network of routes and specified timetable of flights.



Objective of FAP: **Minimize** the combined costs of "spill" (= rejected payload demand) and aircraft operating costs over the entire airline fleet and network.



Payload Spill (larger aircraft)





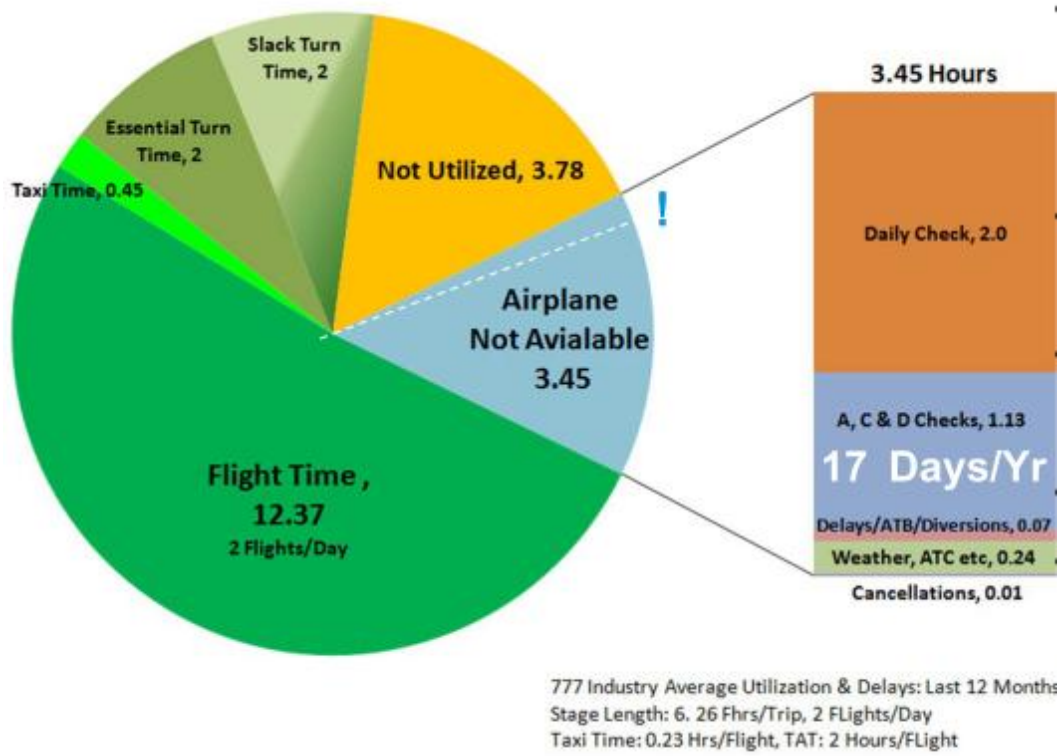
Operating Costs (smaller aircraft)

…subject to: constraints such as minimum ground time, maintenance requirements, number of aircraft by type available, aircraft rotation constraints (= ensure feasible aircraft cycles, balance of aircraft inflow/outflow at each airport)

Source: Belobaba (2009)

Airline, MRO, & Manufacturer Partnership

Enhancing aircraft availability



Source: Boeing (2013)

More reliable airplanes: less delays

- Schedule reliability: from 98% to 99.4% (777)

Rapid parts delivery

- < 24-hour delivery time

Operations center

- 7/24 rapid response

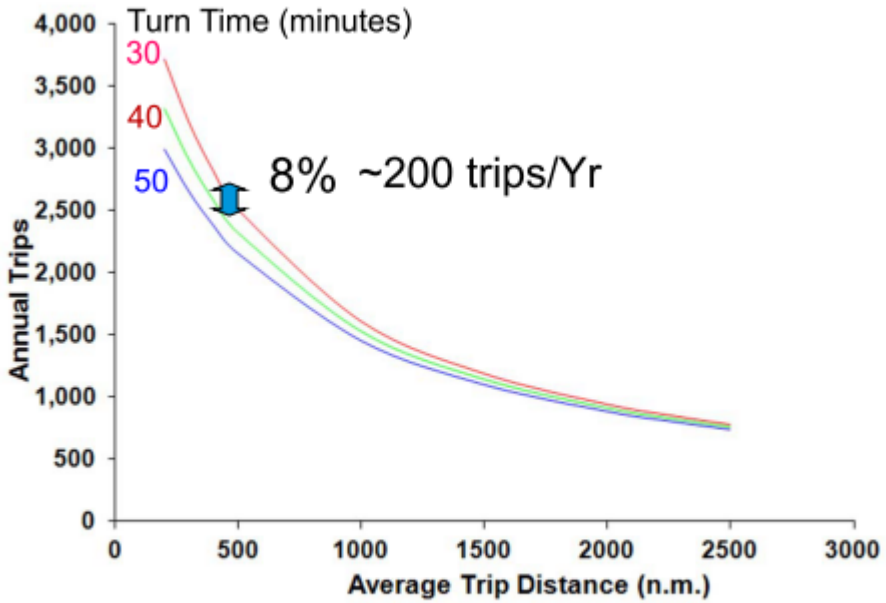
Daily checks tasks reduced

Check Interval Optimization

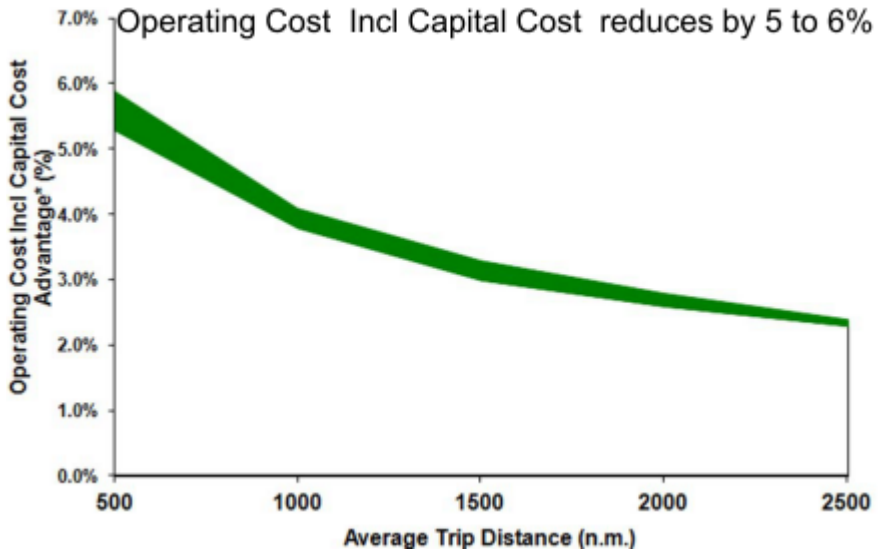
- A-Check: from 600 Fhrs to 1,000 Fhrs
- C-Check: from 2 years to 3 years
- Days-out-of-service: 17 to 12 days

What an Airline can do to improve Aircraft Availability

- Reduce turn-around hours
- Reduce flight slack time
- Consider more routes and red-eyes
- Plan efficient operations



Reducing Turn Time By 10 Minutes Improves Airplane Utilization By 8% @ 500 nm Trip Length



Source: Boeing (2013)

What an MRO can do to improve Aircraft Availability

- Optimize check intervals in line with airline operations
- Reduce airplane delays
- Schedule airplane checks in lean operational times
- Reduce airplane time in hangars
  - More manpower
  - Outsourcing
  - Ready availability of spares and parts

➔ Each hour enhanced utilization = 1 to 4 Mio.$ p.a. profit opportunity!

Source: Boeing (2013)